

Projection-Based First-Order Optimization for Constrained Quaternion Attitude Maneuvers

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Abstract— Deriving optimal attitude maneuvers for agile spacecraft is challenging due to the nonlinearity of quaternion kinematics and the strong nonconvexity of pointing constraints. Sequential Convex Programming (SCP) addresses these issues through repeated linearization, but its solutions can be sensitive to initialization and computationally demanding for repeated replanning. Direct NLP solvers such as IPOPT avoid repeated convexification, but they can still exhibit convergence sensitivity and timing variability due to their monolithic interior-point structure. This paper proposes a first-order, projection-based method in which nonlinear dynamics and cone-type orientation constraints are handled through modular projection operators within an Alternating Direction Method of Multipliers (ADMM) framework. The proposed approach (i) reformulates the attitude optimal control problem as an affine dynamics-consensus system augmented with auxiliary variables, (ii) derives analytical geometric projections for single keep-in/keep-out pointing cone constraints, (iii) handles simultaneous multi-cone constraints through cyclic/local projection refinement, and (iv) enforces nonlinear dynamics through low-dimensional Levenberg–Marquardt graph projections, thereby avoiding repeated horizon-wide convexification. In addition, a conditional local stationarity analysis is provided for the resulting inexact nonconvex ADMM scheme. The analysis characterizes the approximation errors of the multi-cone attitude pro-

jection and the nonlinear dynamics graph projection using residual-based local error bounds, and establishes stationary convergence under local regularity, boundedness, Kurdyka–Łojasiewicz, and summable projection-error assumptions. Numerical experiments on representative low-Earth orbit reorientation scenarios compare the proposed method with SCP and IPOPT baselines. The evaluation includes nominal constrained trajectory generation, randomized Monte Carlo simulations, perturbed trajectory tracking, receding-horizon MPC under uncertainty, hyperparameter sensitivity, and horizon-length scaling. The proposed solver achieves reliable keep-in/keep-out constraint satisfaction and competitive terminal accuracy, is substantially faster than the SCP baseline in the tested scenarios, and provides more consistent closed-loop feasibility than the IPOPT baseline in the receding-horizon study, although IPOPT remains faster in average runtime when it converges. The horizon-scaling and timing-breakdown results further show that the numerical nonlinear dynamics projection remains a small fixed-dimensional local solve over the tested horizon range. These results indicate that the proposed framework can provide efficient and reliable constraint handling for constrained spacecraft attitude guidance problems involving nonconvex constraints.

Index Terms— Agile spacecraft maneuvers, First-order methods, Keep-in/Keep-out constraints, Nonconvex optimal control, Projection-based algorithms

I. INTRODUCTION

Spacecraft operating in low Earth orbit (LEO) must execute rapid, high-fidelity attitude maneuvers while obeying strict power budgets and precise pointing requirements. Typical mission profiles demand: (i) sub-second to few-second payload reorientation, (ii) sub-arcminute line-of-sight accuracy, and (iii) compliance with *keep-in* (mandatory) or *keep-out* (forbidden) cones defined in the body frame to protect sensitive instruments and maintain communication links [1]. The inherent nonlinearity of quaternion kinematics and strong nonconvexity of orientation constraints render such optimal control problems particularly challenging.

Classical methods for spacecraft attitude regulation are primarily based on nonlinear state-feedback controllers [2], [3], [4], [5], [6], [7], often formulated using quaternion kinematics [8], [9]. These approaches are well-studied and widely applied in operational systems. Observer designs on the rotation group $SO(3)$ have also been proposed to enhance robustness against measurement uncertainty and model mismatch [10].

To address geometric or safety-related constraints, more recent strategies incorporate artificial potential functions (APFs) [11], [12], [13], [14], which enable avoidance of undesirable regions while maintaining stability. In parallel, control barrier functions (CBFs) have been developed to enforce state and safety constraints in real-time control implementations [15], [16]. These methods complement Lyapunov-based frameworks and have been increasingly considered in aerospace applications where formal constraint satisfaction is required. However, these methods typically do not account for global performance metrics and may face difficulties when applied to problems with highly nonconvex constraints or strict timing requirements.

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Optimization-based methods have thus gained prominence. Classical convex control design [17], [18] is well-understood, but direct convexification of quaternion dynamics and cone constraints is nontrivial. Sequential Convex Programming (SCP) methods address this by iteratively linearizing and convexifying the constraints and dynamics [19], [20], [21], [22]. Pioneering work in SCP for aerospace includes convex approaches to Mars powered descent [23], [24], with further extensions incorporating state-triggered constraints [25], [26]. While SCP has demonstrated effectiveness in numerous scenarios, its convergence is highly sensitive to the choice of hyperparameters and the quality of the initial solution guess, potentially leading to increased computational demands or unreliable performance in certain cases [27], [28], [29]. Interior-point solvers such as IPOPT [30] have been used for trajectory optimization [31], [32], though their performance may severely degrade under strong nonconvexities or poor initialization. Recent surveys and frameworks have further advanced the landscape of trajectory optimization by integrating robustness, scalability, and real-time implementation goals [33].

First-order methods with projection-based feasibility enforcement have emerged as promising alternatives for solving nonlinear constrained control problems. This approach introduces a mechanism for alternating between gradient-based updates and projections onto the set of admissible trajectories governed by nonlinear dynamics, thereby avoiding repeated linearization and accommodating general nonconvex constraints [34], [35], [36]. Subsequent developments include distributed implementation [37], neural projection operators [38] and projection in the Koopman lifted hyperplanes [39], [40], which enhance scalability and computational efficiency.

In this paper, we develop a projection-based first-order optimization framework for spacecraft attitude maneuver planning with quaternion dynamics and nonconvex keep-in/keep-out orientation constraints. The affine dynamics and consensus terms are handled through a structured least-squares primal update, while nonlinear and nonconvex feasibility requirements are enforced through modular auxiliary projections. Single-cone attitude constraints are handled by analytical geometric projections, simultaneous multi-cone constraints are addressed using cyclic projection and local refinement, and the exponential-map dynamics graph is projected through a low-dimensional numerical least-squares procedure. We further characterize the local errors induced by these inexact projections and establish a conditional local stationarity result for the resulting nonconvex ADMM scheme under standard regularity, boundedness, and summable-error assumptions.

The main contributions of this paper are summarized as follows.

- *Projection-based ADMM formulation for constrained quaternion maneuvers:* We reformulate the constrained quaternion maneuver problem as a dynamics-consensus optimization problem whose primal step is

a structured least-squares update and whose nonlinear constraints are separated into modular projection subproblems.

- *Geometry-preserving discretization and projection operators:* Quaternion kinematics are discretized using a norm-preserving exponential-map update, while unit-quaternion and keep-in/keep-out pointing constraints are enforced through analytical single-cone projections and cyclic multi-cone refinement.
- *Projection accuracy and conditional stationarity analysis:* We quantify the residual errors of the multi-cone attitude projection and Levenberg–Marquardt dynamics projection, and incorporate these errors into a conditional local stationarity result for the proposed nonconvex ADMM scheme.
- *Numerical validation, robustness evaluation, and scalability assessment:* The method is evaluated against SCP and IPOPT baselines in nominal constrained maneuvers, Monte Carlo trajectory generation, perturbed tracking, receding-horizon MPC, hyperparameter sweeps, runtime profiling, and horizon-length scalability tests.

Throughout the numerical experiments, the proposed method achieves reliable pointing-constraint satisfaction, competitive terminal accuracy, and more consistent closed-loop feasibility under the tested uncertainty conditions. It also shows lower sensitivity to the tested penalty parameters than the direct NLP baseline. The horizon-length timing study further indicates smooth computational scaling with respect to the discretization size. Together with the conditional stationarity analysis, these results support the empirical convergence behavior, scalability, and replanning suitability of the proposed framework in the tested constrained attitude maneuver scenarios.

The remainder of this paper is organized as follows. Section II reviews the notation, quaternion representation, and first-order optimization background. Section III formulates the constrained quaternion attitude maneuver planning problem. Section IV presents the proposed projection-based ADMM framework, including conditional stationarity analysis. Section V reports numerical comparisons with baselines. Section VI concludes the paper.

II. PRELIMINARIES

A. Notations

We adopt the block vector convention $(a, b) = [a^T \ b^T]^T$ for any vectors a and b . Using this notation, a trajectory vector is written as $x = (x_1, x_2, \dots, x_N)$, where each x_k denotes the state at time index k .

For any vector $\omega = (\omega_x, \omega_y, \omega_z) \in \mathbb{R}^3$, we define the corresponding skew-symmetric matrix:

$$S(\omega) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix},$$

which satisfies $S(\omega)v = \omega \times v$ for any $v \in \mathbb{R}^3$. We also define the quaternion kinematics matrix:

$$\Omega(\omega) = \begin{bmatrix} 0 & \omega^T \\ -\omega & S(\omega) \end{bmatrix}.$$

B. Quaternion Kinematics and Rotation Maps

Euler angles are a conventional representation for rigid body orientation but suffer from singularities (gimbal lock) at specific configurations, which can impair numerical stability in control and planning. To avoid these issues, we adopt unit quaternions, which provide a globally non-singular and compact representation of three-dimensional rotations using four parameters.

The quaternion algebra $\mathbb{H}$ is identified with $\mathbb{R}^4$ and denotes the set of quaternions:

$$\mathbb{H} = \{q = q_0 + q_1i + q_2j + q_3k \mid q_i \in \mathbb{R}\},$$

where q_0 is the scalar part and $q_v = (q_1, q_2, q_3) \in \mathbb{R}^3$ is the vector part. A unit quaternion satisfies $\|q\| = 1$ and lies on the unit 3-sphere:

$$\mathbb{S}^3 = \{q \in \mathbb{H} \mid \|q\| = 1\}.$$

A rotation by angle θ about a unit vector $e \in \mathbb{R}^3$ is represented by the quaternion, $q = (\cos \frac{\theta}{2}, e \sin \frac{\theta}{2})$, which lies on $\mathbb{S}^3$ by construction.

The corresponding rotation matrix $R(q)$ mapping inertial-frame vectors into the body frame is:

$$R(q) = (q_0^2 - \|q_v\|^2)I_3 + 2q_0S(q_v) + 2q_vq_v^T.$$

C. First-Order Optimization

First-order methods address constrained optimization by combining gradient steps with proximal or projection operations [41]. Their low per-iteration cost and ability to exploit problem structure make them well-suited for embedded and large-scale applications.

A prominent example is the Alternating Direction Method of Multipliers (ADMM), which introduces auxiliary variables to decompose complex problems and iteratively updates primal and dual variables through alternating minimization [42].

Consider a constrained optimization problem of the form:

$$\underset{x \in \mathcal{C}}{\text{minimize}} \quad g(x),$$

which can be reformulated as:

$$\begin{aligned} &\underset{x, z}{\text{minimize}} \quad g(x) + I_{\mathcal{C}}(z) \\ &\text{subject to} \quad x = z, \end{aligned}$$

where $I_{\mathcal{C}}(\cdot)$ is the indicator function of the set $\mathcal{C}$.

ADMM solves this via the augmented Lagrangian:

$$\mathcal{L}_\rho(x, z, u) = g(x) + I_{\mathcal{C}}(z) + \frac{\rho}{2}\|x - z + u\|^2,$$

with the following update rules:

$$\begin{aligned} x^{(k+1)} &= \text{prox}_{g/\rho}(z^{(k)} - u^{(k)}), \\ z^{(k+1)} &= \Pi_{\mathcal{C}}(x^{(k+1)} + u^{(k)}), \\ u^{(k+1)} &= u^{(k)} + x^{(k+1)} - z^{(k+1)}, \end{aligned}$$

where $\Pi_{\mathcal{C}}$ denotes the Euclidean projection onto the set $\mathcal{C}$.

Convergence is typically assessed using the primal and dual residuals:

$$r_p^{(k+1)} = \|x^{(k+1)} - z^{(k+1)}\|, \quad r_d^{(k+1)} = \rho\|z^{(k+1)} - z^{(k)}\|,$$

and the algorithm terminates once both residuals fall below predefined tolerances.

While convergence guarantees hold when $\mathcal{C}$ is convex, many real-world problems involve nonconvexities (e.g., nonlinear dynamics or conic constraints). In such cases, projection steps may not be nonexpansive, and global convergence cannot be ensured. Nevertheless, first-order projection methods have demonstrated practical effectiveness in many nonconvex settings [35], [36].

Recent studies [43], [44], [45] provide partial theoretical support for the behavior of these methods under nonconvexity and inexact updates.

D. Nonlinear Control via First-Order Optimization

We consider a general discrete-time nonlinear optimal control problem, where the objective is to determine a control sequence that minimizes a performance cost while satisfying dynamics and operational constraints.

The problem can be formulated as:

$$\begin{aligned} &\underset{x, u}{\text{minimize}} \quad g(u_0, \dots, u_{N-1}, x_1, \dots, x_N) \\ &\text{subject to} \quad x_N = x_{\text{des}}, \\ &\quad \quad \quad x_{t+1} = f_0(x_t, u_t), \\ &\quad \quad \quad (x_{t+1}, u_t) \in \mathcal{C}_t^{\text{con}}, \quad \forall t \in \{0, \dots, N-1\}, \end{aligned}$$

where x_t and u_t denote the state and control input at time step t , and $f_0(x_t, u_t)$ represents the nonlinear system dynamics. The cost function $g(\cdot)$ may capture control effort, tracking error, or other performance criteria. The initial condition is fixed as $x_0 = x_{\text{init}}$, and each constraint set $\mathcal{C}_t^{\text{con}} = \mathcal{C}_t^{\mathcal{X}} \times \mathcal{C}_t^{\mathcal{U}}$ encodes allowable regions for state and control.

To handle the nonlinear dynamics constraint, we assume a control-affine structure:

$$f_0(x_t, u_t) = A_t x_t + B_t u_t + f(x_t),$$

and introduce auxiliary variables $s_t = f(x_t)$ to isolate the nonlinear term. Define the set

$$\mathcal{C}_t^{\text{dyn}} = \{(s_t, x_t) \mid s_t = f(x_t)\},$$

so that the nonlinear relationship is expressed implicitly. This separation facilitates compatibility with ADMM, enabling constraint enforcement via modular projection steps.

The reformulated problem becomes:

$$\begin{aligned} &\underset{x, u, s}{\text{minimize}} \quad g(u_0, \dots, u_{N-1}, x_1, \dots, x_N) \\ &\text{subject to} \quad x_N = x_{\text{des}}, \\ &\quad \quad \quad x_{t+1} = A_t x_t + B_t u_t + s_t, \\ &\quad \quad \quad (s_t, x_t) \in \mathcal{C}_t^{\text{dyn}}, \\ &\quad \quad \quad (x_{t+1}, u_t) \in \mathcal{C}_t^{\text{con}}, \end{aligned} \tag{1}$$

for all $t \in \{0, \dots, N-1\}$.

This formulation decouples the linear update rule from the nonlinear component, allowing each constraint to be handled independently via projection.

Problem (1) is compatible with an ADMM scheme, in which constraints such as C_t^{dyn} and C_t^{con} are enforced through separate projection operations.

By reformulating the problem as

$$\begin{aligned} & \underset{y,z}{\text{minimize}} && g(y) + I_C(z) \\ & \text{subject to} && Cy = d + z, \end{aligned}$$

one can alternate between updating y via a structured quadratic subproblem, projecting z onto the constraint set $\mathcal{C}$, and updating the dual variable accordingly.

This modular decomposition supports efficient implementation and allows for the integration of approximate or learned projection operators. As a result, the solver can accurately enforce dynamics while flexibly handling a wide range of operational constraints [36], [38], [39].

III. PROBLEM STATEMENT: QUATERNION ATTITUDE MANEUVER PLANNING

This section formalizes the optimal attitude maneuver planning problem for spacecraft equipped with reaction wheel systems. The objective is to compute control torque inputs that steer the spacecraft between designated orientations, while satisfying dynamic constraints, actuator limits, and mission-specific pointing requirements. The quaternion-based kinematic and dynamic models are first introduced, followed by orientation constraints expressed as quadratic conditions, and finally the complete optimal control problem is formulated.

A. Quaternion Attitude Dynamics

The quaternion-based rigid body dynamics [46] are given by

$$\begin{aligned} \dot{q} &= -\frac{1}{2}\Omega(\omega)q, \\ \dot{\omega} &= -J^{-1}S(\omega)J\omega + J^{-1}u, \end{aligned}$$

where $q \in \mathbb{R}^4$ is the unit quaternion, $\omega \in \mathbb{R}^3$ is the angular velocity vector in the body frame, $u \in \mathbb{R}^3$ is the applied control torque, and $J \in \mathbb{R}^{3 \times 3}$ is the positive definite moment of inertia matrix. The matrix $\Omega(\omega) \in \mathbb{R}^{4 \times 4}$ encodes the quaternion kinematics, and $S(\omega)$ is the skew-symmetric matrix corresponding to the cross product with ω .

REMARK 1 (Unit-quaternion invariance in continuous time) *From $\dot{q} = -\frac{1}{2}\Omega(\omega)q$ with $\Omega(\omega)^T = -\Omega(\omega)$, it follows that*

$$\frac{d}{dt}\|q(t)\|^2 = 2q(t)^T\dot{q}(t) = -q(t)^T\Omega(\omega(t))q(t) = 0,$$

so $\|q(t)\|$ is an invariant of the continuous dynamics. In particular, if $\|q(t_0)\| = 1$ then $\|q(t)\| = 1$ for all t .

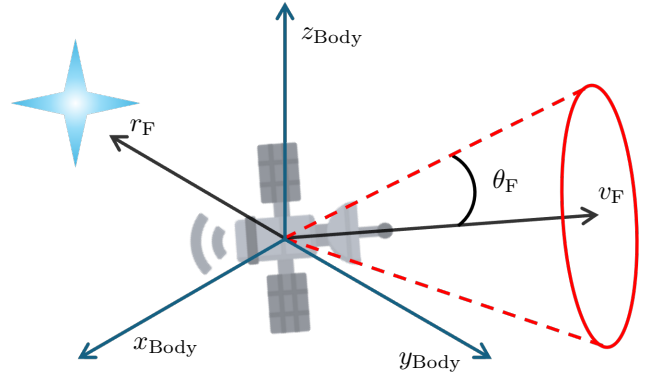


Fig. 1. Illustration of a keep-out (forbidden) cone constraint, where the body-frame direction v_F must avoid pointing within an angle θ_F of the inertial vector r_F , typically representing hazardous or occluded regions.

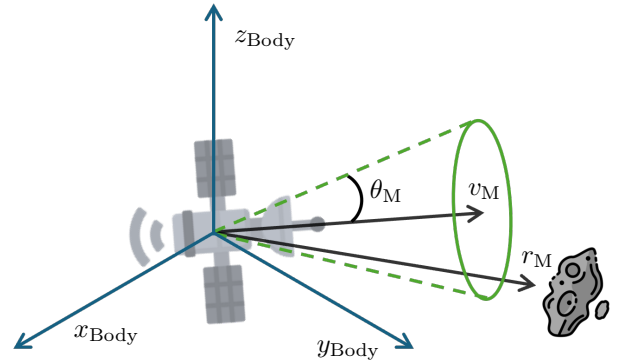


Fig. 2. Illustration of a keep-in (mandatory) cone constraint, where the body-frame direction v_M is required to maintain alignment within an angular deviation θ_M from the inertial reference vector r_M .

B. Pointing-Cone Constraints as Quadratic Forms

Mission-specific pointing requirements often impose constraints on the spacecraft's attitude during maneuvers. These are typically categorized as either keep-out constraints, which define regions to be avoided (Figure 1), or keep-in constraints, which specify regions the attitude must remain within (Figure 2). Both types of conditions are naturally represented as conic regions in the body frame.

Let $v \in \mathbb{R}^3$ be a fixed direction in the body frame (e.g., aligned with a sensor axis), and let $r \in \mathbb{R}^3$ be a reference direction in the inertial frame (e.g., pointing toward the Sun or a target star). The spacecraft orientation is described by the rotation matrix $R(q)$, determined by the unit quaternion $q \in \mathbb{R}^4$, which maps inertial vectors into the body frame.

The cosine of the angle between v and r is given by $v^T R(q)r$, which admits a closed-form expression in terms of quaternion components:

$$v^T R(q)r = v^T \left((q_0^2 - q_v^T q_v) I_3 + 2q_0 S(q_v) + 2q_v q_v^T \right) r,$$

where q_0 is the scalar part and $q_v \in \mathbb{R}^3$ is the vector part of the quaternion.

Using this, orientation constraints can be formulated as quadratic forms in q .

A keep-out (forbidden) constraint requires that the angle between the body-fixed direction v_F and inertial direction r_F exceed a specified threshold θ_F :

$$v_F^T R(q) r_F \leq \cos \theta_F \iff q^T C(v_F, r_F, \theta_F) q \leq 0,$$

Conversely, a keep-in (mandatory) constraint requires that the angle between v_M and r_M remain below a threshold θ_M :

$$v_M^T R(q) r_M \geq \cos \theta_M \iff q^T C(v_M, r_M, \theta_M) q \geq 0,$$

where the matrix $C(v, r, \theta) \in \mathbb{R}^{4 \times 4}$ is defined as:

$$C(v, r, \theta) = \begin{bmatrix} v^T r - \cos \theta & v^T S(r)^T \\ S(r)v & vr^T + rv^T - (v^T r + \cos \theta) I_3 \end{bmatrix}.$$

For notational convenience, we define $C_F = C(v_F, r_F, \theta_F)$ and $C_M = C(v_M, r_M, \theta_M)$ to represent the constraint matrices for the keep-out and keep-in conditions, respectively.

C. Constrained Quaternion Maneuver Optimization

The spacecraft maneuver planning problem is formulated as an optimal control problem. The objective is to compute a control torque profile $u(t)$ that steers the spacecraft from an initial condition to a desired terminal state, while minimizing control effort and satisfying dynamic, physical, and mission-specific cone constraints.

$$\underset{q(\cdot), \omega(\cdot), u(\cdot)}{\text{minimize}} \int_{t_0}^{t_f} \|u(t)\|^2 dt \quad (2a)$$

$$\text{subject to} \quad \begin{bmatrix} \dot{q}(t) \\ \dot{\omega}(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}\Omega(\omega)q(t) \\ -J^{-1}S(\omega)J\omega(t) \end{bmatrix} + \begin{bmatrix} 0 \\ J^{-1} \end{bmatrix} u(t), \quad (2b)$$

$$q(t)^T q(t) = 1, \quad (2c)$$

$$q(t)^T C_F q(t) \leq 0, \quad (2d)$$

$$q(t)^T C_M q(t) \geq 0, \quad (2e)$$

$$\|u(t)\|_\infty \leq u_{\max}, \quad \|\omega(t)\|_\infty \leq \omega_{\max}, \quad (2f)$$

$$q(t_0) = q_{\text{init}}, \quad \omega(t_0) = \omega_{\text{init}}, \quad (2g)$$

$$q(t_f) = q_{\text{des}}, \quad \omega(t_f) = \omega_{\text{des}}. \quad (2h)$$

Here, constraint (2b) defines the nonlinear quaternion dynamics, while (2c) enforces unit-quaternion normalization. The quadratic cone constraints (2d)–(2e) ensure compliance with keep-out and keep-in pointing requirements. Bounds in (2f) reflect actuator and operational limitations, and (2g)–(2h) specify the initial and terminal boundary conditions.

The problem is inherently nonconvex due to the nonlinear dynamics and quadratic orientation constraints, which limits the applicability of conventional convex optimization techniques without approximation. The proposed method addresses this by employing a projection-based first-order optimization framework that accommodates nonconvexities while reducing computational overhead relative to repeated convexification-based approaches.

IV. FIRST-ORDER OPTIMIZATION FRAMEWORK FOR ATTITUDE MANEUVER PLANNING

This section presents a first-order optimization framework for solving the constrained spacecraft attitude maneuver problem introduced in Section III. Due to the inherent nonconvexities in quaternion dynamics and orientation constraints, classical convex optimization techniques are inadequate. Our approach leverages the Alternating Direction Method of Multipliers (ADMM) with projection-based techniques capable of efficiently handling nonconvex constraint sets.

A. Discretization and Dynamics Decomposition

We discretize the continuous-time problem in (2) using a fixed time step Δt . Let $q_t \in \mathbb{R}^4$, $\omega_t \in \mathbb{R}^3$, and $u_t \in \mathbb{R}^3$ denote the quaternion, body angular velocity, and control torque at time index t , respectively. Define the state vector

$$x_t := \begin{bmatrix} q_t \\ \omega_t \end{bmatrix} \in \mathbb{R}^7.$$

The continuous-time rigid-body attitude dynamics in (2) can be written in control-affine form as

$$\dot{x}(t) = B_c u(t) + f_c(x(t)), \quad (3)$$

where

$$B_c := \begin{bmatrix} 0_{4 \times 3} \\ J^{-1} \end{bmatrix}, \quad f_c(q, \omega) := \begin{bmatrix} -\frac{1}{2}\Omega(\omega)q \\ -J^{-1}S(\omega)J\omega \end{bmatrix}.$$

A direct forward-Euler discretization of (3) gives

$$x_{t+1} = x_t + \Delta t B_c u_t + \Delta t f_c(x_t). \quad (4)$$

Introducing an auxiliary variable ν_t to isolate the nonlinear term yields the affine-plus-graph representation

$$x_{t+1} = x_t + \Delta t B_c u_t + \Delta t \nu_t, \quad \nu_t = f_c(x_t).$$

This decomposition is useful for ADMM because the inter-sample coupling is affine, while the nonlinear dynamics are confined to the graph constraint $\nu_t = f_c(x_t)$. However, the forward-Euler update does not preserve the unit-quaternion geometry at the discrete level.

REMARK 2 (Quaternion norm drift under Euler propagation) *Consider the quaternion component of (4):*

$$q_{t+1}^E = \left(I_4 - \frac{\Delta t}{2} \Omega(\omega_t) \right) q_t. \quad (5)$$

Since $\Omega(\omega_t)^T = -\Omega(\omega_t)$, we have

$$\begin{aligned} \|q_{t+1}^E\|^2 &= q_t^T \left(I_4 - \frac{\Delta t}{2} \Omega(\omega_t) \right)^T \left(I_4 - \frac{\Delta t}{2} \Omega(\omega_t) \right) q_t \\ &= \|q_t\|^2 + \frac{\Delta t^2}{4} q_t^T \Omega(\omega_t)^T \Omega(\omega_t) q_t. \end{aligned}$$

For the quaternion kinematics matrix used in this paper, $\Omega(\omega_t)^T \Omega(\omega_t) = \|\omega_t\|^2 I_4$. Therefore,

$$\|q_{t+1}^E\|^2 = \left(1 + \frac{\Delta t^2}{4} \|\omega_t\|^2 \right) \|q_t\|^2.$$

Thus, even if $\|q_t\| = 1$, the Euler-propagated quaternion generally satisfies $\|q_{t+1}^E\| > 1$ unless $\omega_t = 0$. For bounded

angular rates $\|\omega_t\| \leq \omega_{\max}$ over a fixed horizon $T = N\Delta t$, the accumulated norm-square drift is bounded by

$$\begin{aligned} \|q_N^E\|^2 &\leq \|q_0\|^2 \left(1 + \frac{\Delta t^2}{4} \omega_{\max}^2\right)^N \\ &\leq \|q_0\|^2 \exp\left(\frac{T\Delta t}{4} \omega_{\max}^2\right). \end{aligned}$$

Hence the accumulated drift vanishes as $\Delta t \rightarrow 0$, but it is not identically zero for finite time steps. This drift can introduce artificial inconsistency between the discrete dynamics graph and the unit-quaternion constraint.

To remove this discretization-induced norm drift, the quaternion kinematics are discretized using the exponential map. Assuming zero-order hold on ω_t over the interval $[t_t, t_t + \Delta t]$, the exact discrete update of

$$\dot{q}(t) = -\frac{1}{2}\Omega(\omega_t)q(t)$$

is

$$q_{t+1} = \Phi_q(\omega_t, \Delta t)q_t, \quad (6)$$

where the transition matrix $\Phi_q(\omega_t, \Delta t)$ is given by the matrix exponential as,

$$\Phi_q(\omega_t, \Delta t) := \exp\left(-\frac{\Delta t}{2}\Omega(\omega_t)\right).$$

Let

$$\alpha_t := \frac{\Delta t}{2}\|\omega_t\|.$$

Using $\Omega(\omega_t)^2 = -\|\omega_t\|^2 I_4$, the transition matrix admits the closed form

$$\Phi_q(\omega_t, \Delta t) = \cos(\alpha_t)I_4 - \frac{\Delta t}{2}\text{sinc}(\alpha_t)\Omega(\omega_t),$$

where

$$\text{sinc}(\alpha) := \frac{\sin \alpha}{\alpha}, \quad \text{sinc}(0) := 1.$$

The removable singularity at $\alpha = 0$ is evaluated using the analytic extension of $\text{sinc}(\alpha)$.

REMARK 3 (Norm preservation of the exponential-map update) *Because $\Omega(\omega_t)^T = -\Omega(\omega_t)$, the transition matrix $\Phi_q(\omega_t, \Delta t)$ is orthogonal:*

$$\Phi_q(\omega_t, \Delta t)^T \Phi_q(\omega_t, \Delta t) = I_4.$$

Consequently,

$$\|q_{t+1}\| = \|\Phi_q(\omega_t, \Delta t)q_t\| = \|q_t\|.$$

Thus, if $q_t \in \mathbb{S}^3$, then the exponential-map propagation satisfies $q_{t+1} \in \mathbb{S}^3$ up to numerical roundoff. Unlike the Euler update in (5), the exponential-map update does not introduce artificial quaternion norm drift at the discrete level.

The angular-velocity dynamics are discretized by a first-order update:

$$\omega_{t+1} = \omega_t + \Delta t (-J^{-1}S(\omega_t)J\omega_t + J^{-1}u_t). \quad (7)$$

Combining (6) and (7), the discrete dynamics retain the affine-plus-graph structure

$$x_{t+1} = x_t + \Delta t B_c u_t + \Delta t \nu_t, \quad (8)$$

but the nonlinear graph variable is now defined by

$$\nu_t = f_{\exp}(q_t, \omega_t), \quad (9)$$

where

$$f_{\exp}(q_t, \omega_t) := \begin{bmatrix} \frac{(\Phi_q(\omega_t, \Delta t) - I_4)q_t}{\Delta t} \\ -J^{-1}S(\omega_t)J\omega_t \end{bmatrix}. \quad (10)$$

The first component of $f_{\exp}$ is the exponential-map quaternion increment divided by Δt . As $\Delta t \rightarrow 0$, it satisfies

$$\frac{(\Phi_q(\omega_t, \Delta t) - I_4)q_t}{\Delta t} = -\frac{1}{2}\Omega(\omega_t)q_t + \mathcal{O}(\Delta t),$$

so (10) is consistent with the continuous-time dynamics while preserving the unit-quaternion geometry under finite-step propagation.

Using (8)–(10), the discretized optimal control problem is written as

$$\underset{q, \omega, \nu, u}{\text{minimize}} \quad \sum_{t=0}^{N-1} \|u_t\|^2 \quad (11a)$$

$$\text{subject to} \quad \begin{bmatrix} q_{t+1} \\ \omega_{t+1} \end{bmatrix} = \begin{bmatrix} q_t \\ \omega_t \end{bmatrix} + \Delta t B_c u_t + \Delta t \nu_t, \quad (11b)$$

$$\nu_t = f_{\exp}(q_t, \omega_t), \quad (11c)$$

$$q_{t+1}^T q_{t+1} = 1, \quad (11d)$$

$$q_{t+1}^T C_F q_{t+1} \leq 0, \quad (11e)$$

$$q_{t+1}^T C_M q_{t+1} \geq 0, \quad (11f)$$

$$\|u_t\|_{\infty} \leq u_{\max}, \quad \|\omega_{t+1}\|_{\infty} \leq \omega_{\max}, \quad (11g)$$

$$q_0 = q_{\text{init}}, \quad \omega_0 = \omega_{\text{init}}, \quad (11h)$$

$$q_N = q_{\text{des}}, \quad \omega_N = \omega_{\text{des}}, \quad (11i)$$

$$\forall t \in \{0, \dots, N-1\}.$$

The decision variables over the horizon are $q = (q_1, \dots, q_N)$, $\omega = (\omega_1, \dots, \omega_N)$, $\nu = (\nu_0, \dots, \nu_{N-1})$, and $u = (u_0, \dots, u_{N-1})$, while q_0 and ω_0 are fixed by the initial condition. The combination of (11b) and (11c) separates the affine temporal coupling from the nonlinear dynamics graph. The corresponding exponential-map dynamics graph is

$$\mathcal{C}_{\exp}^{\nu} := \prod_{t=0}^{N-1} \{(q_t, \omega_t, \nu_t) \mid \nu_t = f_{\exp}(q_t, \omega_t)\}.$$

On any bounded angular-rate domain, $f_{\exp}$ is smooth and real analytic because the sinc term is evaluated through its analytic extension at the origin. This graph structure is used in the ADMM splitting to enforce nonlinear dynamical consistency through the projection onto $\mathcal{C}_{\exp}^{\nu}$.

REMARK 4 (Unit-quaternion enforcement in the discrete ADMM problem) *The exponential-map update preserves the norm of a quaternion propagated through (6). However, the primal update of ADMM is performed in the ambient Euclidean space and may generate intermediate quaternion variables that do not lie exactly on $\mathbb{S}^3$. Therefore, the unit-quaternion constraint is explicitly enforced in the auxiliary projection associated with*

$$\mathcal{C}^q = \{q \in \mathbb{R}^4 \mid \|q\| = 1, q^T C_F q \leq 0, q^T C_M q \geq 0\}.$$

B. ADMM Splitting with Auxiliary Variables

To solve the discretized optimal control problem using ADMM, we reformulate the objective and constraints into a structure suitable for alternating minimization. ADMM operates by iteratively minimizing the augmented Lagrangian with respect to the primal variables, followed by proximal updates that enforce constraint satisfaction via projections.

The terminal conditions on the state variables in (11i) are relaxed and incorporated as soft penalty term in the objective function with weight parameter λ_N .

To enable constraint decomposition, we introduce auxiliary variables z^q, z^ω, z^ν , and z^u , which are duplicates of the primal variables q, ω, ν , and u , respectively. Additionally, split variables $z^{q\nu}$ and $z^{\omega\nu}$ are introduced to decouple the nonlinear dynamical consistency condition $\nu = f_{\text{exp}}(q, \omega)$, resulting in a combined auxiliary variable $z = (z^q, z^\omega, z^\nu, z^u, z^{q\nu}, z^{\omega\nu})$. Each auxiliary variable is constrained to lie in a predefined set, and the associated indicator function is included in the objective to enforce membership in that set.

This reformulation leads to an ADMM objective composed of a quadratic cost and multiple indicator terms, enabling efficient, structure-exploiting updates for each variable block:

$$\begin{aligned} \underset{q, \omega, \nu, u, z}{\text{minimize}} \quad & \sum_{t=0}^{N-1} \|u_t\|^2 + \lambda_N (\|q_N - q_{\text{des}}\|^2 + \|\omega_N - \omega_{\text{des}}\|^2) \\ & + \sum_{t=0}^{N-1} \left(I_{C^u}(z_t^u) + I_{C^\nu}(z_t^\nu, z_t^{q\nu}, z_t^{\omega\nu}) \right. \\ & \quad \left. + I_{C^q}(z_{t+1}^q) + I_{C^\omega}(z_{t+1}^\omega) \right) \\ \text{subject to} \quad & G(q, \omega, \nu, u) = 0, \\ & u_t - z_t^u = 0, \quad \nu_t - z_t^\nu = 0, \\ & q_t - z_t^{q\nu} = 0, \quad \omega_t - z_t^{\omega\nu} = 0, \\ & q_{t+1} - z_{t+1}^q = 0, \quad \omega_{t+1} - z_{t+1}^\omega = 0, \\ & \forall t \in \{0, \dots, N-1\}, \end{aligned}$$

where the constraint sets are defined as:

$$C^\nu = \{(q, \omega, \nu) \mid \nu = f_{\text{exp}}(q, \omega)\}, \quad (12a)$$

$$C^\omega = \{\omega \mid \|\omega\|_\infty \leq \omega_{\text{max}}\}, \quad (12b)$$

$$C^u = \{u \mid \|u\|_\infty \leq u_{\text{max}}\}, \quad (12c)$$

$$C^q = \{q \mid \|q\| = 1, q^T C_{\text{F}} q \leq 0, q^T C_{\text{M}} q \geq 0\}. \quad (12d)$$

Here, $G(q, \omega, \nu, u)$ denotes the stacked affine dynamics residual in (11b), with q_0 and ω_0 fixed by (11h). Each constraint set C^l enforces feasibility for a specific subsystem: C^q ensures quaternion normalization and cone compliance; C^ν enforces nonlinear dynamical consistency $\nu_t = f_{\text{exp}}(q_t, \omega_t)$; and C^u, C^ω impose actuator and rate limits.

C. Augmented Lagrangian and Optimization Structure

The augmented Lagrangian for this formulation, shown in (13), incorporates all constraints using dual variables and quadratic penalties to encourage feasibility.

The optimization proceeds by alternating the following updates:

- *Primal variable update*: Minimize the augmented Lagrangian with respect to the primal variables (q, ω, ν, u) , keeping auxiliary and dual variables fixed.
- *Auxiliary variable update*: Project the auxiliary variables $z = (z^q, z^\omega, z^\nu, z^u, z^{q\nu}, z^{\omega\nu})$ onto their respective constraint sets $\mathcal{C}$, enforcing both convex and nonconvex feasibility.
- *Dual variable update*: Perform standard dual ascent to update the Lagrange multipliers $w = (w^q, w^\omega, w^\nu, w^u, w^{q\nu}, w^{\omega\nu}, w^G)$, encouraging consensus between primal and auxiliary variables.

The augmented Lagrangian is presented in two complementary forms: the expanded form (13a) and the compact form (13b). The expanded form shows explicit summations over time steps, clarifying the contributions of each constraint and penalty term. This is useful for implementation and interpretation, especially in analyzing temporal coupling introduced by dynamics and constraint duplications.

The compact form instead aggregates the variables into block vectors (e.g., $q = (q_1, \dots, q_N)$), emphasizing the overall structure of the problem as a block-separable augmented Lagrangian.

This formulation naturally supports ADMM-based decomposition. Because the indicator functions and quadratic penalties are separable across time and variable types, projection steps onto sets such as C^u, C^ω, C^q , and C^ν can be performed in parallel across time indices. This separability improves computational efficiency and may be advantageous for resource-limited guidance computation. In the following sections, we leverage this modularity to design a projection-based solver. Each constraint set is handled using specialized projection operators: some convex and closed-form, others nonconvex but computationally tractable via geometric or analytical methods.

D. Primal Variable Update: Regularized Least-Squares Solve

The primal variable update step in (14) corresponds to solving a regularized multi-objective least-squares problem for (q, ω, ν, u) . This minimization aggregates the dynamics residual, control effort, and consensus penalties across all variable types. The resulting subproblem is convex and can be solved efficiently using sparse linear algebra or custom solvers that exploit problem structure.

E. Auxiliary Variable Update: Projection Operators

The auxiliary variable update step in ADMM is performed using proximal operators. Each variable in the

$$\mathcal{L}_\rho(q, \omega, \nu, u, z, w) = \sum_{t=0}^{N-1} \left(\|u_t\|^2 + I_{C^u}(z_t^u) + I_{C^\nu}(z_t^\nu, z_t^{q\nu}, z_t^{\omega\nu}) + I_{C^q}(z_{t+1}^q) + I_{C^\omega}(z_{t+1}^\omega) \right) \quad (13a)$$

$$\begin{aligned} & + \lambda_N \left(\|q_N - q_{\text{des}}\|^2 + \|\omega_N - \omega_{\text{des}}\|^2 \right) + \frac{\rho}{2} \left\{ \|G(q, \omega, \nu, u) + w^G\|^2 \right. \\ & + \sum_{t=0}^{N-1} \left(\|u_t - z_t^u + w_t^u\|^2 + \|\nu_t - z_t^\nu + w_t^{\nu\nu}\|^2 + \|q_t - z_t^{q\nu} + w_t^{q\nu}\|^2 \right. \\ & \left. \left. + \|\omega_t - z_t^{\omega\nu} + w_t^{\omega\nu}\|^2 + \|q_{t+1} - z_{t+1}^q + w_{t+1}^q\|^2 + \|\omega_{t+1} - z_{t+1}^\omega + w_{t+1}^\omega\|^2 \right) \right\} \\ & = \|u\|^2 + \lambda_N \left(\|q_N - q_{\text{des}}\|^2 + \|\omega_N - \omega_{\text{des}}\|^2 \right) \\ & + \sum_{t=0}^{N-1} \left(I_{C^u}(z_t^u) + I_{C^\nu}(z_t^\nu, z_t^{q\nu}, z_t^{\omega\nu}) + I_{C^q}(z_{t+1}^q) + I_{C^\omega}(z_{t+1}^\omega) \right) \\ & + \frac{\rho}{2} \left\{ \|G(q, \omega, \nu, u) + w^G\|^2 + \|u - z^u + w^u\|^2 + \|\nu - z^\nu + w^{\nu\nu}\|^2 \right. \\ & \left. + \|q - z^{q\nu} + w^{q\nu}\|^2 + \|\omega - z^{\omega\nu} + w^{\omega\nu}\|^2 + \|q - z^q + w^q\|^2 + \|\omega - z^\omega + w^\omega\|^2 \right\} \end{aligned} \quad (13b)$$

Primal Variable Update

$$(q, \omega, \nu, u)^{(k+1)} = \arg \min_{q, \omega, \nu, u} \left\{ \|u\|^2 + \lambda_N \left(\|q_N - q_{\text{des}}\|^2 + \|\omega_N - \omega_{\text{des}}\|^2 \right) + \frac{\rho}{2} \left(\|G(q, \omega, \nu, u) + w^{G,(k)}\|^2 \right. \right. \quad (14)$$

$$\begin{aligned} & \left. + \|u - z^{u,(k)} + w^{u,(k)}\|^2 + \|\nu - z^{\nu,(k)} + w^{\nu,(k)}\|^2 + \|q - z^{q\nu,(k)} + w^{q\nu,(k)}\|^2 \right. \\ & \left. + \|\omega - z^{\omega\nu,(k)} + w^{\omega\nu,(k)}\|^2 + \|q - z^{q,(k)} + w^{q,(k)}\|^2 + \|\omega - z^{\omega,(k)} + w^{\omega,(k)}\|^2 \right) \right\} \end{aligned}$$

vector z is updated by projecting onto its corresponding constraint set, as defined in (12). The overall constraint set $\mathcal{C}$ is the intersection of these sets, including both convex and nonconvex subsets.

For convex sets (e.g., input and rate constraints), standard Euclidean projections are used. For nonconvex sets (e.g., nonlinear dynamics constraints), projection is handled using direct geometric projection or iterative projection, which efficiently enforces feasibility without requiring full convex reformulation.

 1. Update z^q via analytical projection onto pointing cones

The set $\mathcal{C}^q$ encodes the unit-quaternion constraint together with cone-type orientation constraints, including keep-in and keep-out pointing requirements. Unit-quaternion normalization is explicitly enforced in the projection step. The exponential-map discretization preserves the unit-quaternion manifold at the discrete level, as discussed in Remark 3. Therefore, the normalization projection is consistent with the nonlinear dynamics graph projection and does not introduce an artificial residual floor. Therefore, we perform the update

$$z_{q,t}^{(k+1)} = \Pi_{C^q}(q_t^{(k+1)} + w_t^{q,(k)}),$$

enforcing both normalization and the cone constraints.

In order to enforce the keep-in (mandatory) and keep-out (forbidden) visibility constraints, solving the following

optimization problem (16) is required.

$$\text{minimize}_{q \in \mathbb{S}^3} \|q - \bar{q}\|^2 \quad (16a)$$

$$\text{subject to } q^T C_M q \geq 0, \quad (\text{or } q^T C_F q \leq 0,) \quad (16b)$$

where the objective minimizes deviation from the given quaternion $\bar{q}$, and the constraints enforce quadratic cone bounds.

Note that the problem is nonconvex due to the indefiniteness of the corresponding quadratic constraint matrices C_M and C_F . However, rather than solving this nonconvex problem directly, we adopt an efficient analytical approximation based on geometric reasoning:

LEMMA 1 (Geometric projection onto a single cone on $\mathbb{S}^3$) *Let $\bar{q} \in \mathbb{S}^3$ be a unit quaternion (current attitude), $v \in \mathbb{R}^3$ a fixed body-axis, and $r \in \mathbb{R}^3$ a fixed inertial vector. Define the body-frame inertial vector $s = R(\bar{q})r$, the rotation axis $e = (s \times v)/\|s \times v\|$, and the present angle $\bar{\theta} = \arccos(v^T s)$.*

(a) Keep-in cone (for $v^T R(q)r \geq \cos \theta_M$): If $\bar{\theta} \leq \theta_M$, the optimal update is $q^ = \bar{q}$. If $\bar{\theta} > \theta_M$, set*

$$\delta = \bar{\theta} - \theta_M,$$

and form the corrective quaternion $q_d = (\cos \frac{\delta}{2}, e \sin \frac{\delta}{2})$. Then the projected quaternion

$$q^* = q_d \otimes \bar{q}$$

$$\begin{aligned}
 z_t^{q,(k+1)} &= \arg \min_{z_t^q} \left\{ I_{C^q}(z_t^q) + \frac{\rho}{2} \|q_t^{(k+1)} - z_t^q + w_t^{q,(k)}\|^2 \right\} = \Pi_{C^q}(q_t^{(k+1)} + w_t^{q,(k)}) = \Pi_{C^q}(\bar{q}_t) \\
 z_t^{\omega,(k+1)} &= \arg \min_{z_t^\omega} \left\{ I_{C^\omega}(z_t^\omega) + \frac{\rho}{2} \|\omega_t^{(k+1)} - z_t^\omega + w_t^{\omega,(k)}\|^2 \right\} = \Pi_{C^\omega}(\omega_t^{(k+1)} + w_t^{\omega,(k)}) = \Pi_{C^\omega}(\bar{\omega}_t) \\
 z_t^u,(k+1) &= \arg \min_{z_t^u} \left\{ I_{C^u}(z_t^u) + \frac{\rho}{2} \|u_t^{(k+1)} - z_t^u + w_t^{u,(k)}\|^2 \right\} = \Pi_{C^u}(u_t^{(k+1)} + w_t^{u,(k)}) = \Pi_{C^u}(\bar{u}_t) \\
 \begin{bmatrix} z_t^{q\nu,(k+1)} \\ z_t^{\omega\nu,(k+1)} \\ z_t^{\nu,(k+1)} \end{bmatrix} &= \arg \min_{(z_t^{q\nu}, z_t^{\omega\nu}, z_t^{\nu})} \left\{ I_{C^\nu} \left(\begin{bmatrix} z_t^{q\nu} \\ z_t^{\omega\nu} \\ z_t^{\nu} \end{bmatrix} \right) + \frac{\rho}{2} \left\| \begin{bmatrix} q_t^{(k+1)} - z_t^{q\nu} + w_t^{q\nu,(k)} \\ \omega_t^{(k+1)} - z_t^{\omega\nu} + w_t^{\omega\nu,(k)} \\ \nu_t^{(k+1)} - z_t^{\nu} + w_t^{\nu,(k)} \end{bmatrix} \right\|^2 \right\} \\
 &= \Pi_{C^\nu} \left(\begin{bmatrix} q_t^{(k+1)} + w_t^{q\nu,(k)} \\ \omega_t^{(k+1)} + w_t^{\omega\nu,(k)} \\ \nu_t^{(k+1)} + w_t^{\nu,(k)} \end{bmatrix} \right) = \Pi_{C^\nu} \left(\begin{bmatrix} \bar{q}_t \\ \bar{\omega}_t \\ \bar{\nu}_t \end{bmatrix} \right)
 \end{aligned}$$

solves

$$\text{minimize } \|q - \bar{q}\|^2 \quad \text{subject to } q^T C_M q \geq 0.$$

(b) *Keep-out cone* (for $v^T R(q)r \leq \cos \theta_F$): If $\bar{\theta} \geq \theta_F$, the optimal update is $q^* = \bar{q}$. If $\bar{\theta} < \theta_F$, set

$$\delta = \bar{\theta} - \theta_F,$$

and define $q_d = (\cos \frac{\delta}{2}, e \sin \frac{\delta}{2})$. Then $q^* = q_d \otimes \bar{q}$ solves

$$\text{minimize } \|q - \bar{q}\|^2 \quad \text{subject to } q^T C_F q \leq 0.$$

In both cases q^* is unique unless $s \parallel v$ (in which case any $e \perp v$ yields the same objective), and $\|q^*\| = 1$.

Proof:

For unit quaternions, the Euclidean distance and geodesic distance on $\mathbb{S}^3$ are monotonically related: for $q, \bar{q} \in \mathbb{S}^3$,

$$\|q - \bar{q}\|^2 = 2(1 - |\langle q, \bar{q} \rangle|) = 2(1 - \cos(\Delta/2)),$$

where Δ is the principal relative rotation angle. Thus minimizing $\|q - \bar{q}\|^2$ over $\mathbb{S}^3$ is equivalent to minimizing Δ .

Recall that $s = R(\bar{q})r$. A left-multiplication by $q_d = (\cos \frac{\alpha}{2}, e \sin \frac{\alpha}{2})$ rotates s about the axis e by angle α inside the body frame. Constraining $e \parallel (s \times v)$ changes only the angle between s and v while achieving the *smallest* α required to meet the inequality with equality ($\bar{\theta} \mapsto \theta_M$ for keep-in or $\bar{\theta} \mapsto \theta_F$ for keep-out). Any other axis would need a strictly larger α to restore feasibility. Since the resulting attitude change is exactly $\Delta = \alpha$, $q_d \otimes \bar{q}$ minimizes the spherical (hence Euclidean-on- $\mathbb{S}^3$) distance among feasible unit quaternions. Uniqueness follows except in the degenerate case $s \parallel v$, where any axis orthogonal to v yields the same minimal α . ■

2. Update (z^ω, z^u) via projection onto ℓ_∞ -norm boxes

The constraint sets C^ω and C^u impose uniform upper bounds on the infinity norms of the control torque and angular velocity vectors, respectively. Explicitly, these sets are defined as:

$$C^\omega = \{\omega \mid \|\omega\|_\infty \leq \omega_{\max}\}, \quad C^u = \{u \mid \|u\|_\infty \leq u_{\max}\}.$$

The ℓ_∞ -norm constraint defines axis-aligned cubes in $\mathbb{R}^3$, centered at the origin, and it enforces that no

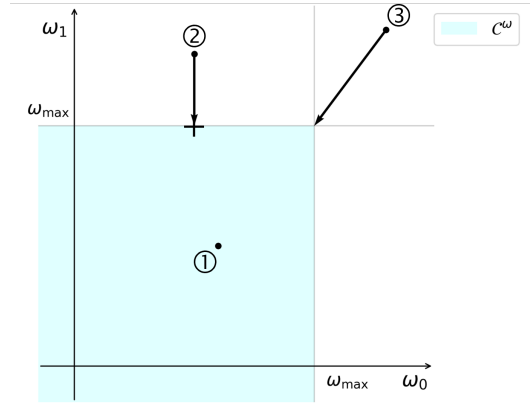


Fig. 3. Projection onto $C^\omega = \{\omega \mid \|\omega\|_\infty \leq \omega_{\max}\}$.

component of the vector exceeds a maximum absolute value, and that this bound is uniform across all directions.

Because both C^u and C^ω are convex and have simple box-shaped geometry, the projection of any vector onto these sets admits a closed-form solution. In particular, each component of the input vector is “saturated” independently to fall within the accepted interval, as shown in Figure 3. This projection is non-expansive, meaning it does not increase the distance between any two projected vectors, and can be performed in parallel across time steps and dimensions.

Applying this principle to the ADMM update step for z^ω and z^u , we obtain the following efficient, closed-form updates:

$$\Pi_{C^\omega}(\bar{\omega}) = \min(|\bar{\omega}|, \omega_{\max}) \cdot \text{sign}(\bar{\omega}),$$

$$\Pi_{C^u}(\bar{u}) = \min(|\bar{u}|, u_{\max}) \cdot \text{sign}(\bar{u}).$$

These projections are performed independently across time steps and dimensions, which can be advantageous for computationally constrained guidance applications.

3. Update $(z^{q\nu}, z^{\omega\nu}, z^\nu)$ via iterative projection onto the dynamics graph

The constraint set C^ν represents the graph of the nonlinear dynamics function $f_{\text{exp}}(x)$, where $x = (q, \omega)$. As shown in Figure 4, the projection $\Pi_{C^\nu}(\bar{x}, \bar{\nu})$ seeks

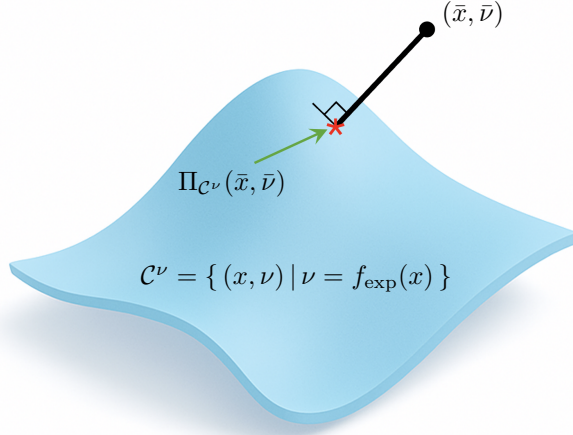


Fig. 4. Orthogonal projection of $(\bar{x}, \bar{\nu})$ onto the nonconvex set $\mathcal{C}^\nu = \{(x, \nu) \mid \nu = f_{\text{exp}}(x)\}$ describing the graph of the nonlinear function $f_{\text{exp}}(\cdot)$.

the closest point on this graph to a given pair $(\bar{x}, \bar{\nu})$. This projection step is inherently nonconvex due to the nonlinearity of $f_{\text{exp}}(x)$.

To perform this projection efficiently, we recast it as a nonlinear least-squares problem:

$$\begin{aligned} & \underset{x, \nu}{\text{minimize}} \quad \|x - \bar{x}\|^2 + \|\nu - \bar{\nu}\|^2 \\ & \text{subject to} \quad \nu = f_{\text{exp}}(x). \end{aligned}$$

where $\bar{x}$ and $\bar{\nu}$ are the current iterate values from the ADMM update.

Substituting the constraint $\nu = f_{\text{exp}}(x)$ into the objective yields the following unconstrained optimization problem.

$$\underset{x}{\text{minimize}} \quad \|x - \bar{x}\|^2 + \|f_{\text{exp}}(x) - \bar{\nu}\|^2$$

This problem is solved numerically using the Levenberg–Marquardt (LM) algorithm [47], [48]. At each inner iteration κ , the function $f_{\text{exp}}(x)$ is approximated by its first-order Taylor expansion around $x^{(\kappa)}$:

$$\begin{aligned} f_{\text{exp}}(x) &\approx \hat{f}_{\text{exp}}(x; x^{(\kappa)}) \\ &= \nabla f_{\text{exp}, x}(x^{(\kappa)})(x - x^{(\kappa)}) + f_{\text{exp}}(x^{(\kappa)}). \end{aligned}$$

The update for $x^{(\kappa)}$ is then obtained by minimizing a regularized least-squares cost:

$$\underset{x}{\text{minimize}} \quad \|x - \bar{x}\|^2 + \|\hat{f}_{\text{exp}}(x; x^{(\kappa)}) - \bar{\nu}\|^2 + \eta \|x - x^{(\kappa)}\|^2$$

where $\eta > 0$ is the LM damping parameter that ensures numerical stability.

The iteration terminates once the update amount falls below a convergence criterion:

$$\|x^{(\kappa+1)} - x^{(\kappa)}\| \leq \varepsilon_{\text{LM}},$$

where ε_{LM} is a predefined threshold for projection accuracy.

In the implementation, the LM iteration at ADMM iteration k and time index t is initialized at the preprojection point

$$x_{t,0}^{(k)} = \bar{x}_t^{(k)} = (q_t^{(k)}, \bar{\omega}_t^{(k)}),$$

which is obtained from the primal update and the scaled dual variable. This initialization is natural for the Euclidean graph projection because the first term in the reduced projection objective penalizes deviation from $\bar{x}_t^{(k)}$. The projected auxiliary variable is then set to

$$(z_t^{q\nu}, z_t^{\omega\nu}, z_t^\nu) = (x_{t,\text{LM}}^{(k)}, f_{\text{exp}}(x_{t,\text{LM}}^{(k)})).$$

Despite the nonconvexity of $\mathcal{C}^\nu$, this projection step remains computationally tractable due to the structure of $f_{\text{exp}}(x)$ and the availability of Jacobians. Moreover, the projection is performed independently for each time step t , enabling parallel execution across the horizon. This parallelizability is beneficial for efficient implementation, allowing complex nonlinear constraints to be handled in a scalable manner.

F. Dual Variable Update

The dual variables are updated using standard dual ascent steps. These updates promote consensus between the primal variables and their associated auxiliary copies. Specifically, the updates at ADMM iteration $k + 1$ are given by:

$$\begin{aligned} w^{G,(k+1)} &= w^{G,(k)} + G(q^{(k+1)}, \omega^{(k+1)}, \nu^{(k+1)}, u^{(k+1)}), \\ w_t^{q,(k+1)} &= w_t^{q,(k)} + q_t^{(k+1)} - z_t^{q,(k+1)}, \\ w_t^{\omega,(k+1)} &= w_t^{\omega,(k)} + \omega_t^{(k+1)} - z_t^{\omega,(k+1)}, \\ w_t^{q\nu,(k+1)} &= w_t^{q\nu,(k)} + q_t^{(k+1)} - z_t^{q\nu,(k+1)}, \\ w_t^{\omega\nu,(k+1)} &= w_t^{\omega\nu,(k)} + \omega_t^{(k+1)} - z_t^{\omega\nu,(k+1)}, \\ w_t^{\nu,(k+1)} &= w_t^{\nu,(k)} + \nu_t^{(k+1)} - z_t^{\nu,(k+1)}, \\ w_t^{u,(k+1)} &= w_t^{u,(k)} + u_t^{(k+1)} - z_t^{u,(k+1)}. \end{aligned}$$

These updates are applied for each time step $t \in \{0, \dots, N - 1\}$ and can be computed in parallel. This decentralized structure is central to the efficiency of the proposed ADMM framework.

G. Residual and Stopping Criteria

Two types of residuals are defined to monitor the convergence of the algorithm: the *primal residual* and the *dual residual*. In the ADMM framework, these residuals measure feasibility violations and the progress of the dual variables, respectively.

The residuals at iteration $k + 1$ are computed as follows:

- Primal residual:

$$p^{(k+1)} = \begin{bmatrix} G(q^{(k+1)}, \omega^{(k+1)}, \nu^{(k+1)}, u^{(k+1)}) \\ q^{(k+1)} - z^{q,(k+1)} \\ \omega^{(k+1)} - z^{\omega,(k+1)} \\ q^{(k+1)} - z^{q\nu,(k+1)} \\ \omega^{(k+1)} - z^{\omega\nu,(k+1)} \\ \nu^{(k+1)} - z^{\nu,(k+1)} \\ u^{(k+1)} - z^{u,(k+1)} \end{bmatrix},$$

- Dual residual:

$$d^{(k+1)} = \begin{bmatrix} z^{q,(k)} - z^{q,(k+1)} \\ z^{\omega,(k)} - z^{\omega,(k+1)} \\ z^{q\nu,(k)} - z^{q\nu,(k+1)} \\ z^{\omega\nu,(k)} - z^{\omega\nu,(k+1)} \\ z^{\nu,(k)} - z^{\nu,(k+1)} \\ z^u,(k) - z^u,(k+1) \end{bmatrix}.$$

The algorithm is considered to have converged when both residuals fall below specified tolerance thresholds:

$$\|p^{(k+1)}\| \leq \varepsilon_{\text{primal}}, \quad \|d^{(k+1)}\| \leq \varepsilon_{\text{dual}}.$$

These stopping conditions are designed to promote feasible and stable iterates, making them useful for efficient trajectory optimization and warm-starting in receding-horizon schemes. The stopping thresholds are selected with reference to the standard absolute and relative ADMM criteria in [42]; the resulting values are then refined through preliminary numerical runs to balance feasibility accuracy and computational cost.

H. Projection Accuracy and ADMM Convergence

Nonconvex ADMM does not have the general convergence guarantees available in the convex setting. Existing results typically require bounded iterates, sufficient descent, regular subproblems, Kurdyka–Łojasiewicz structure, and controlled inexactness [43], [49], [50]. The purpose of this subsection is therefore not to prove global convergence for arbitrary nonconvex instances, but to state a conditional local stationarity result for the proposed projection-based splitting. We first summarize the projection-error models for the multi-cone attitude projection and the Levenberg–Marquardt (LM) dynamics graph projection, and then incorporate these errors into an inexact two-block ADMM analysis.

1. Projection Accuracy

For a single keep-in or keep-out cone, Lemma 1 gives the nearest correction on $\mathbb{S}^3$. With simultaneous cone constraints, the attitude projection at time index t is viewed as the Euclidean projection onto

$$\mathcal{Q}_t := \{q \in \mathbb{R}^4 \mid h(q) = 0, \ g_{\ell,t}(q) \leq 0, \ \ell = 1, \dots, n_c\},$$

where $h(q) := q^T q - 1$. The keep-out and keep-in inequalities are written in the unified form

$$\begin{aligned} g_{F,i,t}(q) &:= q^T C_{F,i,t} q \leq 0, \\ g_{M,j,t}(q) &:= -q^T C_{M,j,t} q \leq 0. \end{aligned}$$

For a preprojection quaternion $\bar{q}_t$, the exact multi-cone projection is

$$\Pi_{\mathcal{Q}_t}(\bar{q}_t) \in \arg \min_{q \in \mathcal{Q}_t} \frac{1}{2} \|q - \bar{q}_t\|^2. \quad (19)$$

Since (19) generally admits no closed-form solution for multiple simultaneous cones, the implemented projection applies the single-cone correction cyclically over the active keep-in and keep-out constraints and is therefore treated

as an approximate projection. For a candidate $\hat{q}_t$, define the projection KKT residual

$$\begin{aligned} \mathcal{R}_t(\hat{q}_t, \bar{q}_t) &:= |h(\hat{q}_t)| + \|[g_t(\hat{q}_t)]_+\|_{\infty} \\ &\quad + \inf_{\lambda_0 \in \mathbb{R}, \lambda \geq 0} \left[\|\hat{q}_t - \bar{q}_t + \lambda_0 \nabla h(\hat{q}_t) + \sum_{\ell=1}^{n_c} \lambda_{\ell} \nabla g_{\ell,t}(\hat{q}_t)\| \right. \\ &\quad \left. + \|\lambda \circ g_t(\hat{q}_t)\|_{\infty} \right], \end{aligned} \quad (20)$$

where $g_t = (g_{1,t}, \dots, g_{n_c,t})$.

The dynamics projection is the projection onto the exponential-map graph

$$\mathcal{C}_{\text{exp},t}^{\nu} := \{(x_t, \nu_t) \mid \nu_t = f_{\text{exp}}(x_t)\}, \quad x_t := (q_t, \omega_t).$$

After eliminating ν_t , the exact graph projection is determined by

$$x_t^* \in \arg \min_x \psi_t(x; \bar{x}_t, \bar{\nu}_t), \quad (21)$$

$$\psi_t(x; \bar{x}_t, \bar{\nu}_t) := \frac{1}{2} \|x - \bar{x}_t\|^2 + \frac{1}{2} \|f_{\text{exp}}(x) - \bar{\nu}_t\|^2. \quad (22)$$

The LM output $\hat{x}_t^{(k)}$ at ADMM iteration k is measured by the reduced first-order residual

$$\mathcal{S}_t(\hat{x}_t^{(k)}; \bar{x}_t^{(k)}, \bar{\nu}_t^{(k)}) := \|\nabla_x \psi_t(\hat{x}_t^{(k)}; \bar{x}_t^{(k)}, \bar{\nu}_t^{(k)})\|, \quad (23)$$

where

$$\nabla_x \psi_t(x_t; \bar{x}_t, \bar{\nu}_t) = x_t - \bar{x}_t + \nabla f_{\text{exp}}(x_t)^T (f_{\text{exp}}(x_t) - \bar{\nu}_t).$$

ASSUMPTION 1 (Regular local projections) *In a neighborhood of the limiting projection points, the following hold. For the cone projection (19), a) $\mathcal{Q}_t$ is nonempty, b) the projection is locally unique, c) the active gradients of h and $g_{\ell,t}$ satisfy LICQ, d) and the second-order sufficient condition holds. For the graph projection (22), e) f_{exp} is continuously differentiable, f) the local minimizer is unique, and g) $\nabla_{xx}^2 \psi_t$ is positive definite at the limiting solution.*

PROPOSITION 1 (Local projection-error bounds) *Suppose Assumption 1 holds. Then, in a sufficiently small regular neighborhood, there exist constants $\kappa_t, \gamma'_t > 0$ such that*

$$\text{dist}(\hat{q}_t, \Pi_{\mathcal{Q}_t}(\bar{q}_t)) \leq \kappa_t \mathcal{R}_t(\hat{q}_t, \bar{q}_t), \quad (24)$$

$$\|(\hat{x}_t, f_{\text{exp}}(\hat{x}_t)) - \Pi_{\mathcal{C}_{\text{exp},t}^{\nu}}(\bar{x}_t, \bar{\nu}_t)\| \leq \gamma'_t \mathcal{S}_t(\hat{x}_t; \bar{x}_t, \bar{\nu}_t). \quad (25)$$

Thus, if the implemented projections satisfy, for all time indices,

$$\mathcal{R}_t(\tilde{\Pi}_{\mathcal{Q}_t}^{(k)}(\bar{q}_t^{(k)}), \bar{q}_t^{(k)}) \leq \varepsilon_k, \quad \mathcal{S}_t(\hat{x}_t^{(k)}; \bar{x}_t^{(k)}, \bar{\nu}_t^{(k)}) \leq \delta_k,$$

and, each projection residual tolerance ε_k, δ_k are summable,

$$\sum_{k=0}^{\infty} \varepsilon_k < \infty, \quad \sum_{k=0}^{\infty} \delta_k < \infty$$

then the stacked attitude and dynamics projection errors satisfy

$$\begin{aligned} e_{q,k} &:= \|\tilde{\Pi}_{\mathcal{Q}}^{(k)}(\bar{q}^{(k)}) - \Pi_{\mathcal{Q}}^{(k)}(\bar{q}^{(k)})\| \leq \bar{\kappa} \sqrt{N} \varepsilon_k, \\ e_{\nu,k} &:= \|\tilde{\Pi}_{\mathcal{C}_{\text{exp}}^{\nu}}^{(k)}(\bar{x}^{(k)}, \bar{\nu}^{(k)}) - \Pi_{\mathcal{C}_{\text{exp}}^{\nu}}^{(k)}(\bar{x}^{(k)}, \bar{\nu}^{(k)})\| \leq \bar{\gamma} \sqrt{N} \delta_k, \end{aligned}$$

where $\bar{\kappa} := \max_t \kappa_t$ and $\bar{\gamma} := \max_t \gamma'_t$. Consequently,

$$\sum_{k=0}^{\infty} \varepsilon_k < \infty, \quad \sum_{k=0}^{\infty} \delta_k < \infty \quad \Rightarrow \quad \sum_{k=0}^{\infty} (e_{q,k} + e_{\nu,k}) < \infty. \quad (26)$$

Proof:

Under LICQ and the second-order sufficient condition, the KKT system of (19) is strongly regular, which gives the local residual error bound in (24). The positive-definite Hessian condition for (22) gives the analogous inverse-function and metric-regularity bound for the reduced graph projection. The stacked bounds follow by summing the per-node errors over the horizon. ■

REMARK 5 (Stacked projection interpretation) *The quantities $e_{q,k}$ and $e_{\nu,k}$ measure the difference between the implemented componentwise projections and the exact product projection $\Pi_Q \times \Pi_{C_{\text{exp}}^\nu}$ in the local regular neighborhood. They are not additional algorithmic variables; they are the projection-error terms used in the inexact ADMM model below.*

REMARK 6 (Loss of local projection regularity) *If the cone intersection is empty, the exact projection is undefined. If the projection is locally nonunique or the active gradients lose rank, the local error bound in Proposition 1 need not apply. These cases mark the boundary of the regular projection neighborhood used in the convergence analysis. If such cases are detected, possible recovery options include a least-violation projection, relaxation of lower-priority pointing constraints, reuse of the previous feasible trajectory, or safe-hold fallback. The numerical examples in this paper were designed with feasible cone configurations and did not require these recovery steps, but the listed mechanisms provide practical extensions for handling degenerate or conflicting constraint configurations when needed.*

2. ADMM Convergence Analysis

Let $y := (q, \omega, \nu, u)$ be the stacked primal trajectory and let $z := (z^q, z^\omega, z^u, z^{\text{dyn}})$ collect the auxiliary variables, with $z^{\text{dyn}} := (z^{\nu}, z^{\omega\nu}, z^\nu)$.

Define

$$\mathcal{C} := \mathcal{C}^q \times \mathcal{C}^\omega \times \mathcal{C}^u \times \mathcal{C}_{\text{exp}}^\nu. \quad (27)$$

The discretized problem can then be written as

$$\begin{aligned} & \underset{y, z}{\text{minimize}} && \ell(y) + I_{\mathcal{C}}(z) \\ & \text{subject to} && \mathcal{A}y + \mathcal{B}z = b, \end{aligned} \quad (28)$$

where

$$\ell(y) = \sum_{t=0}^{N-1} \|u_t\|^2 + \lambda_N (\|q_N - q_{\text{des}}\|^2 + \|\omega_N - \omega_{\text{des}}\|^2), \quad (29)$$

with the terminal penalty omitted when the terminal state is imposed as an affine equality. With the scaled dual variable w , the augmented Lagrangian is

$$\mathcal{L}_\rho(y, z, w) = \ell(y) + I_{\mathcal{C}}(z) + \frac{\rho}{2} \|\mathcal{A}y + \mathcal{B}z - b + w\|^2 - \frac{\rho}{2} \|w\|^2. \quad (30)$$

The ADMM iteration is

$$y^{(k+1)} \in \arg \min_y \mathcal{L}_\rho(y, z^{(k)}, w^{(k)}), \quad (31a)$$

$$z^{(k+1)} \in \arg \min_z \mathcal{L}_\rho(y^{(k+1)}, z, w^{(k)}), \quad (31b)$$

$$w^{(k+1)} = w^{(k)} + \mathcal{A}y^{(k+1)} + \mathcal{B}z^{(k+1)} - b. \quad (31c)$$

The auxiliary update is the product projection

$$\Pi_{\mathcal{C}} = \Pi_{\mathcal{C}^q} \times \Pi_{\mathcal{C}^\omega} \times \Pi_{\mathcal{C}^u} \times \Pi_{\mathcal{C}_{\text{exp}}^\nu}, \quad (32)$$

implemented with the projection errors described in Proposition 1. The primal and dual residuals are

$$\begin{aligned} r^{(k+1)} &:= \mathcal{A}y^{(k+1)} + \mathcal{B}z^{(k+1)} - b, \\ s^{(k+1)} &:= \rho \mathcal{A}^T \mathcal{B} (z^{(k+1)} - z^{(k)}). \end{aligned} \quad (33)$$

Define the total auxiliary projection error

$$\eta_k := e_{q,k} + e_{\nu,k}. \quad (34)$$

By (26), summable projection residuals imply

$$\sum_{k=0}^{\infty} \eta_k < \infty. \quad (35)$$

ASSUMPTION 2 (ADMM regularity and inexactness) *The sequence $\{(y^{(k)}, z^{(k)}, w^{(k)})\}$ remains in a compact regular neighborhood; the least-squares matrix in the y update has full column rank; and the auxiliary projection error η_k satisfies (35).*

The function $\mathcal{L}_\rho$ has the Kurdyka–Łojasiewicz property on the compact regular neighborhoods considered here: ℓ and the affine penalty are quadratic, $\mathcal{C}^u$, $\mathcal{C}^\omega$, and $\mathcal{C}^q$ are described by finitely many box, sphere, and quadratic cone constraints, and $\mathcal{C}_{\text{exp}}^\nu$ is locally described by real analytic equations on the bounded angular-rate domain.

THEOREM 1 (Conditional stationary convergence of the ADMM iterates) *Suppose that Assumptions 1 and 2 hold. Then there exists $\bar{\rho} > 0$ such that, for every $\rho > \bar{\rho}$, the sequence generated by (31) has finite length,*

$$\sum_{k=0}^{\infty} (\|y^{(k+1)} - y^{(k)}\| + \|z^{(k+1)} - z^{(k)}\| + \|w^{(k+1)} - w^{(k)}\|) < \infty. \quad (36)$$

Moreover, $r^{(k)} \rightarrow 0$, $s^{(k)} \rightarrow 0$, and the whole sequence converges to a stationary point (y^*, z^*, w^*) satisfying

$$\begin{aligned} 0 &= \nabla \ell(y^*) + \rho \mathcal{A}^T w^*, \\ 0 &\in N_{\mathcal{C}}(z^*) + \rho \mathcal{B}^T w^*, \\ 0 &= \mathcal{A}y^* + \mathcal{B}z^* - b. \end{aligned} \quad (37)$$

If the active attitude constraints remain regular at the limit, then (37) gives the KKT stationarity system of the discretized attitude maneuver problem.

Proof:

The strongly convex least-squares y update, local Lipschitz behavior of the regular projections, and sufficiently large ρ yield the standard inexact nonconvex ADMM estimates

$$\begin{aligned} & \mathcal{L}_\rho(y^{(k)}, z^{(k)}, w^{(k)}) - \mathcal{L}_\rho(y^{(k+1)}, z^{(k+1)}, w^{(k+1)}) \\ & \geq c_1 (\|y^{(k+1)} - y^{(k)}\|^2 + \|z^{(k+1)} - z^{(k)}\|^2) - \eta_k, \end{aligned}$$

and

$$\|\xi^{(k+1)}\| \leq c_2(\|y^{(k+1)} - y^{(k)}\| + \|z^{(k+1)} - z^{(k)}\|) + \eta_k,$$

where, $\xi^{(k+1)} \in \partial \mathcal{L}_\rho(y^{(k+1)}, z^{(k+1)}, w^{(k+1)})$, for constants $c_1, c_2 > 0$. Since $\sum_k \eta_k < \infty$ and $\mathcal{L}_\rho$ has the Kurdyka–Łojasiewicz property on the compact neighborhood, the usual KL argument gives the finite-length property (36). The dual update gives $w^{(k+1)} - w^{(k)} = r^{(k+1)}$, hence $r^{(k)} \rightarrow 0$. The optimality condition of the y update can be written as

$$0 = \nabla \ell(y^{(k+1)}) + \rho A^T w^{(k+1)} + \rho A^T \mathcal{B}(z^{(k)} - z^{(k+1)}),$$

which implies $s^{(k)} \rightarrow 0$ as $\|z^{(k+1)} - z^{(k)}\| \rightarrow 0$. Passing to the limit in the y - and z -optimality conditions and using the consensus relation gives (37). At a regular attitude point, the normal cone to $\mathcal{C}^q$ is generated by the active unit-norm and cone-constraint gradients, so the split stationarity conditions reduce to the KKT conditions of the discretized problem. ■

REMARK 7 (Numerical projection tolerances) *The sequences $e_{q,k}$ and $e_{v,k}$ represent the ideal projection errors used in the asymptotic analysis. The summability condition in Theorem 1 should be interpreted as a limiting model in which the attitude and dynamics projection tolerances are tightened along the ADMM iterations. The numerical implementation uses fixed tolerances for computational efficiency and therefore does not, by itself, constitute an exact asymptotic realization of the summable-error model. Instead, the reported primal residual, dual residual, cone margins, cone projection residuals, and Levenberg–Marquardt graph-projection residuals quantify the finite-tolerance stationarity and feasibility accuracy achieved by the implemented solver.*

V. NUMERICAL EXPERIMENTS

This section presents numerical simulations for agile fly-by observation missions. The mission is designed to reorient the spacecraft such that its onboard sensor points toward a target precisely at the moment of closest approach. The terminal attitude is defined so that the camera boresight (aligned with the body-frame x -axis) aligns with the line-of-sight (LOS) vector to the target at minimum separation. The terminal angular velocity is chosen to match the angular rate of the LOS vector at this instant. A schematic of the mission geometry is illustrated in Figure 5.

We evaluate two mission scenarios:

- 1) Agile fly-by observation without keep-in or keep-out constraints.
- 2) Agile fly-by observation under both keep-in and keep-out orientation constraints.

To further characterize the robustness and practical applicability of the proposed method, we additionally perform three simulation studies for the constrained fly-by observation mission involving keep-out avoidance.

- 1) Monte Carlo trajectory generation simulation with randomized initial conditions.

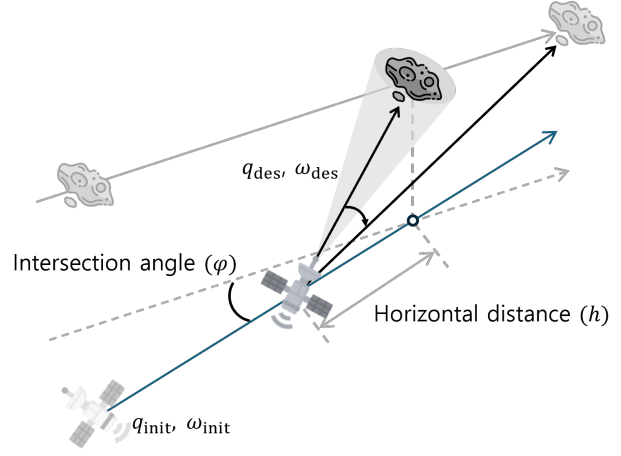


Fig. 5. Agile spacecraft fly-by observation mission.

- 2) Trajectory tracking simulation under model and execution uncertainties.
- 3) Receding-horizon MPC simulation under model and execution uncertainties.

All numerical simulations are compared against two representative trajectory optimization baselines to evaluate the computational performance and solution quality of the proposed method.

The first baseline is a Sequential Convex Programming (SCP) method implemented via `cvxpy` and solved using the `ECOS` SOCP solver [51]. To improve numerical stability, we employed a trust-region update strategy. At each SCP iteration, the convexified subproblem minimizes the penalized objective

$$J_{\text{SCP}} = \lambda_{\text{obj}} f_0(x) + \lambda_{\text{ineq}} \sum_{i=1}^m (f_i(x))_+ + \lambda_{\text{eq}} \sum_{i=1}^p |h_i(x)|,$$

where λ_{ineq} and λ_{eq} are penalty weights for inequality and equality constraint violations, respectively.

Let $J_{\text{true}}^{(k-1)}$ denote the nonlinear penalty cost evaluated at the previous accepted iterate, $\hat{J}^{(k)}$ the objective value of the convexified subproblem, and $J_{\text{true}}^{(k)}$ the nonlinear penalty cost of the candidate solution. The predicted and actual decreases are defined as

$$\Delta \hat{J}^{(k)} = J_{\text{true}}^{(k-1)} - \hat{J}^{(k)}, \quad \Delta J^{(k)} = J_{\text{true}}^{(k-1)} - J_{\text{true}}^{(k)}.$$

The candidate solution is accepted if

$$\Delta J^{(k)} \geq \rho_{\text{scp}} \Delta \hat{J}^{(k)},$$

where ρ_{scp} is the acceptance ratio. If the candidate is accepted, the trust-region radius is expanded; otherwise, it is contracted:

$$r^{(k+1)} = \begin{cases} \gamma_{\text{inc}} r^{(k)}, & \text{if accepted,} \\ \gamma_{\text{dec}} r^{(k)}, & \text{otherwise,} \end{cases}$$

where $\gamma_{\text{inc}} = 1.1$ and $\gamma_{\text{dec}} = 0.5$ typically.

The SCP iteration is terminated when the trust-region radius becomes smaller than the prescribed minimum radius or when the maximum number of SCP iterations is reached:

$$r^{(k)} \leq r_{\min} \quad \text{OR} \quad k \geq k_{\max}.$$

TABLE I
Implementation environment.

Item	Specification
CPU	13th Gen Intel(R) Core(TM) i5-13500
Operating system	Windows 10, version 10.0.26200-SP0
Python	Ver. 3.10.18, Anaconda distribution
NumPy / SciPy	Ver. 1.26.4 / Ver. 1.15.3
cvxpy / CasADi	Ver. 1.7.2 / Ver. 3.7.2

TABLE II
Simulation parameters for Scenario 1.

Parameter	Value	Unit
N	50	–
(t_0, t_f)	(0, 30)	sec
Δt	0.6	sec
J	(40, 50, 55)	kg m ²
q_{init}	(1, 0, 0, 0)	–
q_{des}	(0.7071, 0, 0, -0.7071)	–
ω_{init}	(0, 0, 0)	rad/s
ω_{des}	(0.0, 0.0047, -0.2690)	rad/s
ω_{max}	0.3	rad/s
u_{max}	1.2	N m
(φ, h)	(2, 0)	(deg, m)

The SCP subproblems are solved using the ECOS SOCP solver [51]. For a fair comparison, the penalty weights are tuned for each scenario.

The second baseline is a direct nonlinear programming (NLP) formulation solved using IPOPT through CasADi. The IPOPT baseline solves the same discretized trajectory-optimization problem as in (11). For consistency with the proposed method, the terminal state constraint is relaxed as a quadratic penalty in the objective function, while the remaining path constraints are enforced directly.

All methods are implemented and evaluated on the same computing platform under identical mission scenarios, horizon length, discretization time step, and initial trajectory. The initial trajectory is generated using a PD controller that enforces only the terminal quaternion condition, with gains $K_p = 0.4$ and $K_d = 1.5$. For the proposed method, the auxiliary consensus variables z^i , $i \in \{q, \omega, u, \nu, q_\nu, \omega_\nu\}$, are initialized from the corresponding components of the initial trajectory, and all scaled dual variables w^j , $j \in \{G, q, \omega, u, \nu, q_\nu, \omega_\nu\}$, are initialized to zero.

The hardware and software environment are summarized in Table I.

A. Scenario 1: Unconstrained Fly-By Reorientation

The first scenario evaluates maneuver planning without any conic orientation constraints: only the initial and terminal boundary conditions are enforced.

The final attitude and angular velocity are computed based on the relative geometry at closest approach. Specifically, the terminal quaternion q_{des} is selected to align the camera boresight (body x -axis) with the LOS vector to

TABLE III
Algorithm parameters for Scenario 1.

Proposed	
Stopping criteria	$\varepsilon_{\text{primal}} = 10^{-4}$, $\varepsilon_{\text{dual}} = 10^{-3}$, $\varepsilon_{\text{LM}} = 10^{-3}$
Penalty weights	$(\rho, \lambda_N, \eta) = (1, 20, 10^{-3})$
Max iter.	10000
SCP	
Initial trust region	$r^{(0)} = 1.0$
Stopping criteria	$r_{\text{min}} = 10^{-3}$
Penalty weights	$(\lambda_{\text{obj}}, \lambda_{\text{ineq}}, \lambda_{\text{eq}}) = (2, 1, 2)$
Max iter.	20
IPOPT	
Stopping criteria	$\varepsilon_{\text{IPOPT}} = 10^{-6}$
Penalty weights	$(\lambda_u, \lambda_N) = (1, 20)$
Max iter.	1000

TABLE IV
Optimal control cost, terminal error and computation time for Scenario 1.

	Proposed	SCP	IPOPT
$\ u\ ^2$	22.8098	23.1548	22.8152
$\ x_{t_f} - x_{\text{des}}\ $	0.0030	0.0185	0.0029
Computation time [sec]	0.9534	6.7065	0.1275

the target. The terminal angular velocity ω_{des} matches the instantaneous LOS rate, ensuring smooth tracking across the fly-by.

Simulation parameters and algorithmic parameters for each method for this scenario are listed in Tables II–III. The results are summarized in Table IV and Figures 6–8.

As shown in Table IV, the proposed projection-based method achieves terminal performance comparable to that of the baseline solvers. The terminal error is computed as the deviation between the desired terminal state x_{des} and the final state x_N , obtained by integrating the original nonlinear dynamics using the optimized control inputs. The proposed method achieves the lowest control cost, while maintaining a terminal error nearly identical to the IPOPT result. Compared with SCP, the proposed method provides both lower control cost and substantially smaller terminal error. In terms of computation time, IPOPT is the fastest for this scenario, whereas the proposed method is significantly faster than SCP. These results show that the proposed method achieves solution quality comparable to direct NLP while avoiding the high computational cost observed in the SCP baseline.

B. Scenario 2: Constrained Fly-By Reorientation with Keep-In/Keep-Out Cones

In this scenario, the satellite performs a dual-purpose maneuver: aligning with the observation target while actively avoiding bright celestial objects, such as the Sun, that could damage or saturate its sensors. At each time step, the payload boresight must remain within a specified angular range (θ_2) relative to the target LOS vector

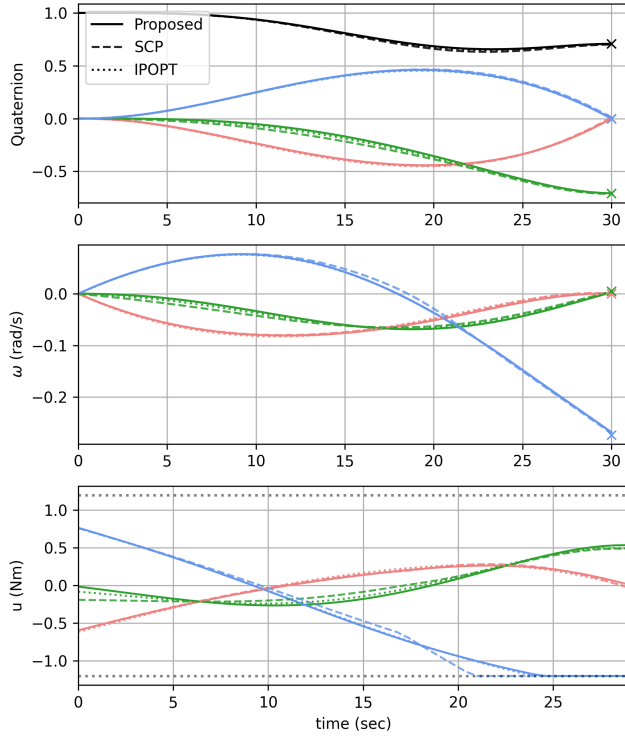


Fig. 6. Scenario 1 spacecraft state and control histories. Top to bottom: quaternion, body-frame angular velocity, and commanded torque. Colors denote q_0 – q_3 in black/red/green/blue and $x/y/z$ axes in red/green/blue.

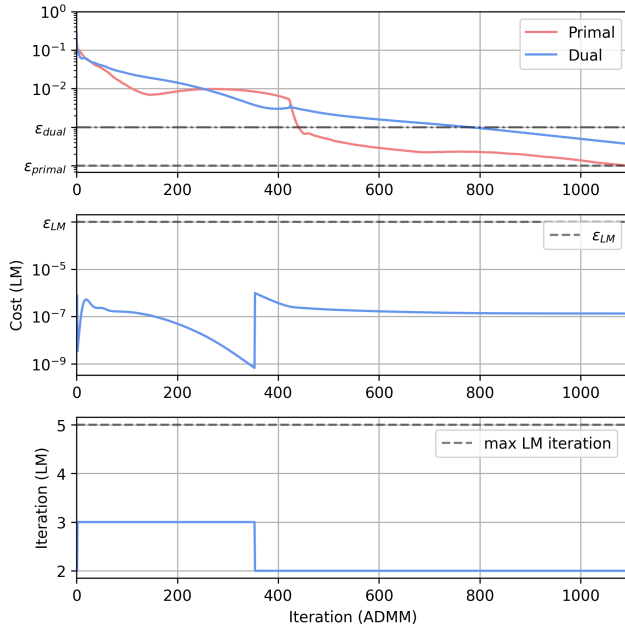


Fig. 7. Convergence behavior of the proposed method for Scenario 1. Primal and dual residuals (top), LM objective value (middle), and LM iteration counts (bottom). The algorithm used termination tolerances of $\varepsilon_{\text{primal}} = 10^{-4}$, $\varepsilon_{\text{dual}} = 10^{-3}$, and $\varepsilon_{\text{LM}} = 10^{-3}$.

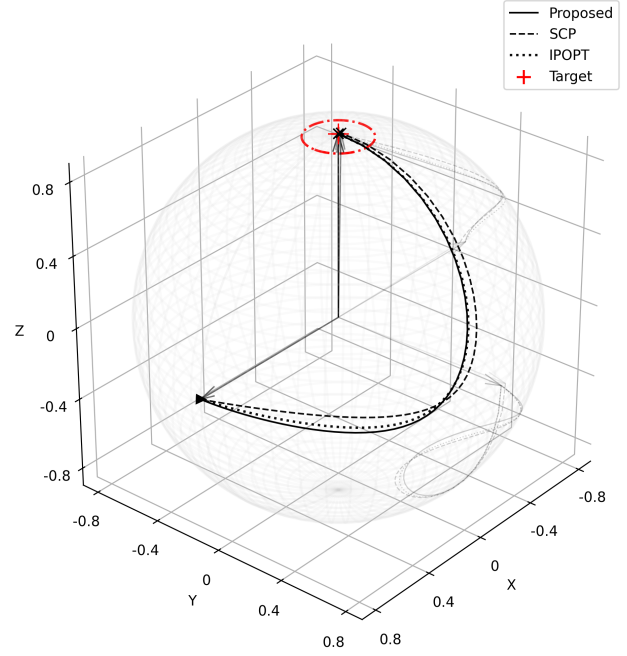


Fig. 8. Attitude maneuver trajectory for Scenario 1. Proposed method (solid) vs. SCP baseline (dashed). Both achieve the desired terminal state under angular velocity constraints.

TABLE V
Simulation parameters for Scenario 2.

Parameter	Value	Unit
N	50	–
(t_0, t_f)	(0, 30)	sec
Δt	0.6	sec
J	(40, 50, 55)	kg m ²
q_{init}	(0.7010, 0, −0.0856, 0.7081)	–
q_{des}	(0.5255, 0.4731, −0.5255, 0.4731)	–
ω_{init}	(0, 0, 0)	rad/s
ω_{des}	(0.0, −0.2670, −0.0328)	rad/s
ω_{max}	0.3	rad/s
u_{max}	1.2	N m
(φ, h)	(2, 0)	(deg, m)
(θ_1, θ_2)	(30, 8)	(deg, deg)
r_1, v_1	(0.4108, 0.7469, 0.5228), (1, 0, 0)	–

(keep-in constraint), while simultaneously maintaining a minimum angular separation (θ_1) from the Sun's LOS vector (keep-out constraint). These angular requirements are encoded as quadratic cone constraints and enforced throughout the maneuver.

Simulation parameters and algorithmic parameters are summarized in Tables V–VI, and the results are shown in Table VII and Figures 9–13.

The proposed method achieves the lowest control cost, which is slightly lower than that of IPOPT and SCP. The terminal error of the proposed method is comparable to that of IPOPT and slightly larger than the SCP result. Compared with Scenario 1, the terminal error increases, which is likely due to the added keep-in and keep-

TABLE VI
Algorithm parameters for Scenario 2.

Proposed	
Stopping criteria	$\varepsilon_{\text{primal}} = 10^{-4}$, $\varepsilon_{\text{dual}} = 10^{-3}$, $\varepsilon_{\text{LM}} = 10^{-3}$
Penalty weights	$(\rho, \lambda_N, \eta) = (1, 10, 10^{-3})$
Max iter.	10000
SCP	
Initial trust region	$r^{(0)} = 10^{-1}$
Stopping criteria	$r_{\min} = 10^{-3}$
Penalty weights	$(\lambda_{\text{obj}}, \lambda_{\text{ineq}}, \lambda_{\text{eq}}) = (10, 20, 20)$
Max iter.	30
IPOPT	
Stopping criteria	$\varepsilon_{\text{IPOPT}} = 10^{-6}$
Penalty weights	$(\lambda_u, \lambda_N) = (1, 10)$
Max iter.	1000

TABLE VII

Optimal control cost, terminal error and computation time for Scenario 2.

	Proposed	SCP	IPOPT
$\ u\ ^2$	15.7673	17.8136	15.7965
$\ x_{t_f} - x_{\text{des}}\ $	0.0215	0.0201	0.0214
Computation time [sec]	0.7870	16.0610	0.2005

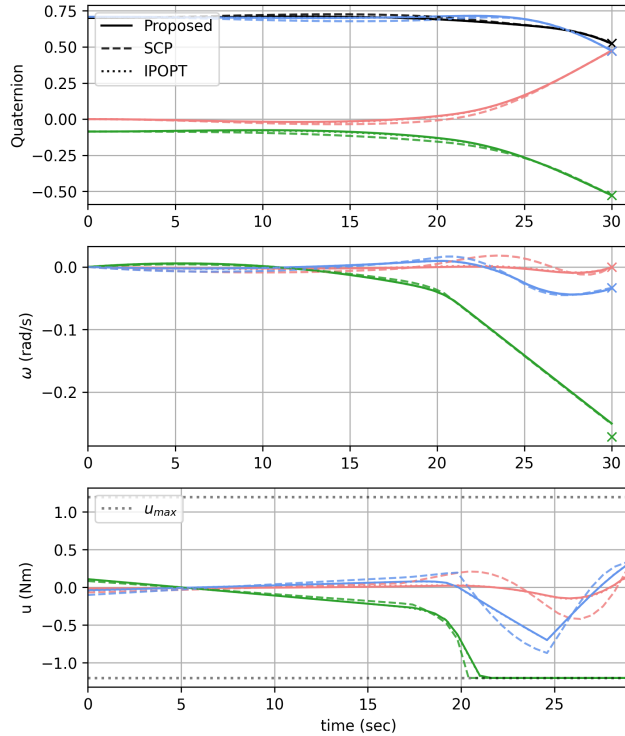


Fig. 9. Scenario 2 state and control histories under keep-in/keep-out constraints. Top to bottom: quaternion, body-frame angular velocity, and commanded torque. Colors denote q_0 – q_3 in black/red/green/blue and $x/y/z$ axes in red/green/blue.

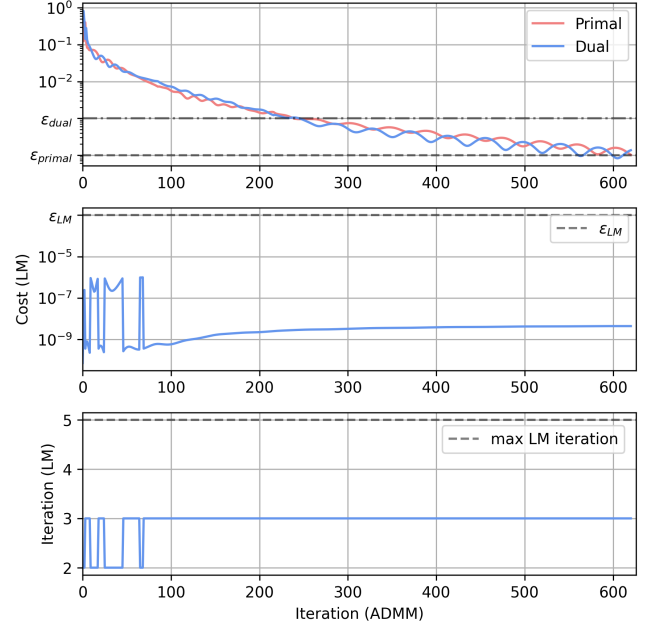


Fig. 10. Convergence behaviors of the proposed method for Scenario 2. Primal and dual residuals (top), LM objective value (middle), and LM iteration counts (bottom). The algorithm used termination tolerances of $\varepsilon_{\text{primal}} = 10^{-4}$, $\varepsilon_{\text{dual}} = 10^{-3}$, $\varepsilon_{\text{LM}} = 10^{-3}$.

out constraints increasing the problem complexity and reducing the relative influence of the penalized terminal constraint. In terms of computation time, IPOPT is the fastest, while the proposed method is substantially faster than SCP. These results indicate that the proposed method achieves competitive solution quality with a much lower computational cost than the SCP baseline. Since the SCP handles the constraints through penalty terms in the objective, its longer computation time is partly due to the additional convexification process. Thus, the computation time advantage of the proposed method and IPOPT should be interpreted in light of this formulation difference.

The residual convergence and optimization progress are shown in Figure 10, demonstrating stable and rapid convergence under the chosen stopping criteria.

To validate constraint satisfaction, Figures 11 and 12 show the trajectories of the Sun and target projected onto the sensor boresight view. These pixel-plane plots are computed from the quaternion state and LOS vectors. As shown, every method effectively maintains visibility of the target within the camera field of view, while keeping the Sun safely outside the protected region with negligible error.

C. Monte Carlo trajectory generation

The Monte Carlo trajectory generation simulation investigates the robustness of the proposed method to variations in the initial condition illustrated in Figure 14. For each trial, the initial state is randomly perturbed around the nominal condition, and the constrained fly-

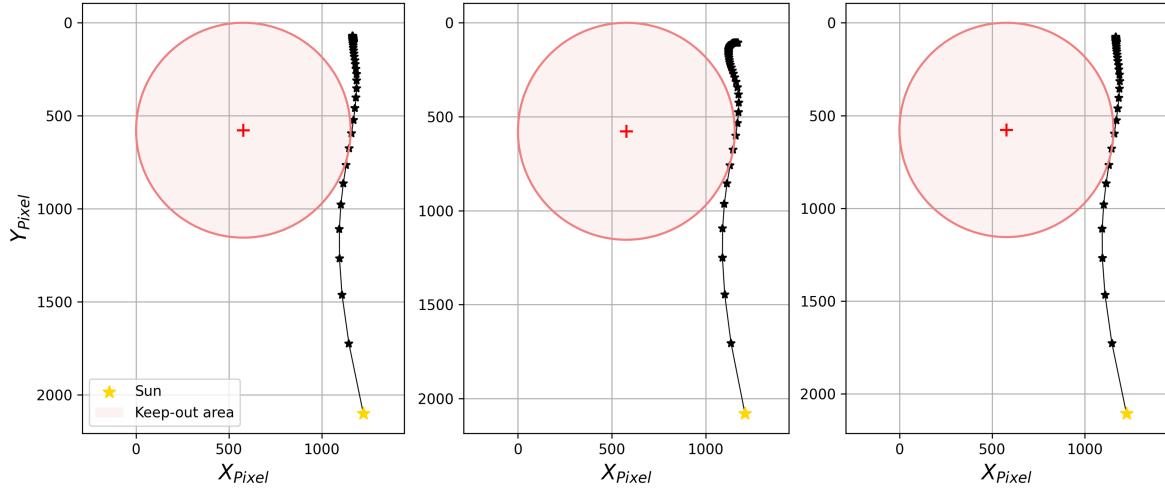


Fig. 11. Sun trajectory on the boresight view. Proposed (left) vs. SCP (middle) vs. IPOPT (right). The proposed method avoids the Sun throughout the maneuver, satisfying the keep-out constraint in Scenario 2.

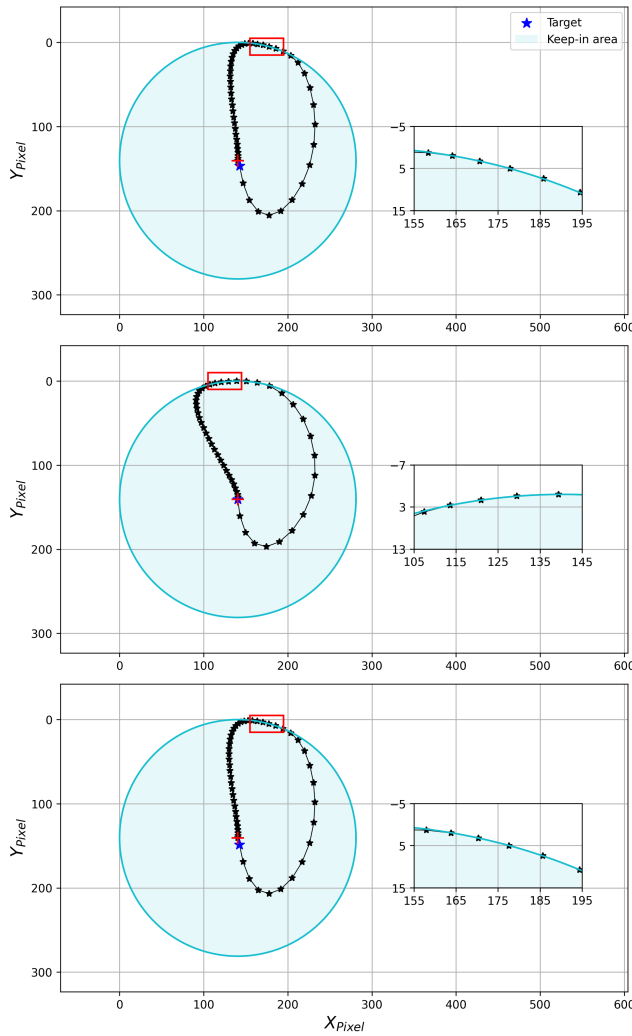


Fig. 12. Target trajectory on the boresight view (Scenario 2). Proposed (top) vs. SCP (middle) vs. IPOPT (bottom). The proposed method maintains continuous visibility within the prescribed angular bound in Scenario 2.

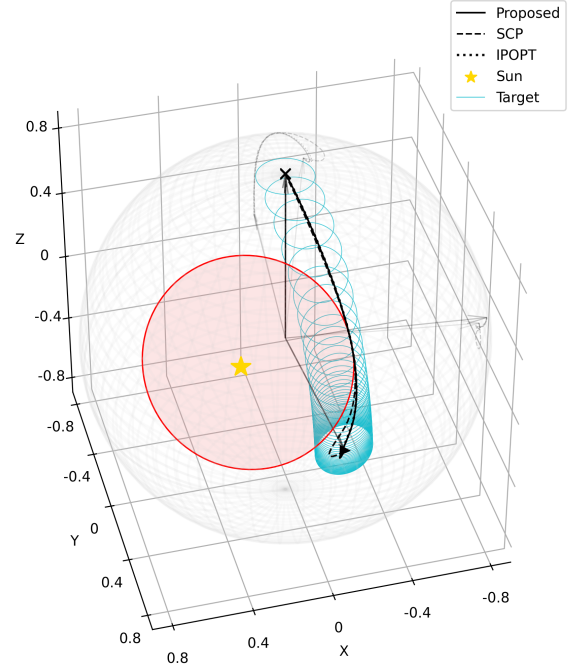


Fig. 13. Attitude maneuver trajectory for Scenario 2. The proposed method (solid) successfully steers toward the target while respecting cone constraints.

by observation trajectory optimization problem is solved independently using the proposed method, SCP, and IPOPT.

All methods use the same mission scenario, horizon length, discretization time step, and terminal objective. After optimization, the computed control sequence is propagated using nonlinear rigid-body dynamics, and the success metrics are evaluated on the propagated trajectory.

A trial is classified as successful if the terminal attitude error, terminal angular-rate error, keep-out violation, control bound, and angular rate bound satisfy the thresholds

TABLE VIII

Success thresholds for Monte Carlo nominal trajectory generation.

Metric	Value	Unit	Metric	Value	Unit
ε_q	2×10^{-1}	deg	ε_ω	2×10^{-2}	rad/s
ε_F	2×10^{-1}	deg	ε_b	1×10^{-6}	–

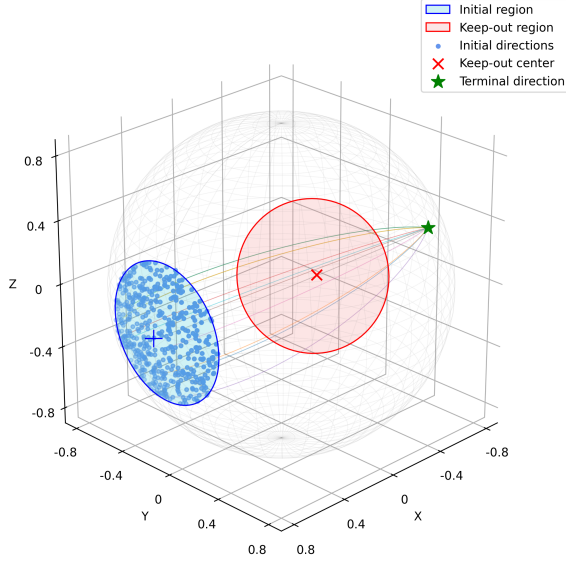


Fig. 14. Monte Carlo planning geometry. The initial pointing region, sampled initial directions, terminal pointing direction, and keep-out region are shown on the unit sphere.

summarized in Table VIII. The Monte Carlo results are used to compare the success rate, computational efficiency, and solution quality of the three methods.

- Terminal attitude error:

$$e_q^f = 2 \arccos(|q_N^T q_{\text{des}}|) \leq \varepsilon_q.$$

- Terminal angular-rate error:

$$e_\omega^f = \|\omega_N - \omega_{\text{des}}\| \leq \varepsilon_\omega.$$

- Maximum keep-out constraint violation:

$$\delta_F^{\max} = \max \{0, -m_F^{\min}\} \leq \varepsilon_F,$$

where

$$m_F^{\min} = \min_{t=0, \dots, N} [\arccos(v_F^T R(q_t) r_F) - \theta_F].$$

- Control bound satisfaction:

$$\max_t \|u_t\|_\infty \leq u_{\max} + \varepsilon_b.$$

- Angular-rate bound satisfaction:

$$\max_t \|\omega_t\|_\infty \leq \omega_{\max} + \varepsilon_b.$$

Tables IX and X summarize the Monte Carlo nominal planning results for randomized initial attitudes. The proposed method achieves the highest success rate of 98.4%, while SCP and IPOPT achieve success rates of 97.8% and 98.0%, respectively. Although SCP yields the smallest mean terminal attitude error, it exhibits a nonzero mean keep-out violation of 0.0746 deg and requires the

largest mean runtime of 4.6582 s. In contrast, the proposed method satisfies the keep-out constraint on average with zero mean violation, while maintaining a moderate mean terminal attitude error of 0.1292 deg.

In terms of computational performance, IPOPT provides the shortest mean runtime of 0.4422 s, but it failed to compute feasible solutions 8 times and reaches the maximum iteration limit of 1000 in some cases. This indicates that the direct NLP formulation can be computationally efficient when it converges, but its convergence reliability is less robust under randomized initial conditions. The proposed method requires a longer mean runtime than IPOPT, but it is substantially faster than SCP and reports no solver failures. Furthermore, the proposed method obtains the lowest mean cost among the three methods, indicating that it provides competitive solution quality while maintaining reliable feasibility and convergence behavior.

D. Trajectory tracking simulation

To assess the executability of the optimized constrained trajectory under model and execution uncertainties, a representative nominal solution is used as the reference trajectory for feedback tracking. The corresponding optimized state and control histories,

$$\{q_t^{\text{ref}}, \omega_t^{\text{ref}}, u_t^{\text{ref}}\}_{t=0}^N \quad (38)$$

are used to define the feedforward and tracking terms.

The true plant is assumed to have an inertia mismatch,

$$J_{\text{true}} = J + \Delta J,$$

where J_{true} remains positive definite. For the diagonal uncertainty case,

$$J_{\text{true}} = \text{diag}(J_1(1 + \delta_{J,1}), J_2(1 + \delta_{J,2}), J_3(1 + \delta_{J,3})),$$

where $|\delta_{J,i}| \leq \bar{\delta}_J$.

The applied torque includes actuator gain error, constant bias, and additive disturbance:

$$\hat{u}_t = \text{sat}_{u_{\max}}(D_u u_t^{\text{cmd}} + u_b + \tau_{d,t}), \quad (39)$$

where

$$D_u = \text{diag}(1 + \delta_{u,1}, 1 + \delta_{u,2}, 1 + \delta_{u,3}).$$

The perturbed plant is propagated from the true state as

$$\begin{aligned} q_{t+1}^{\text{true}} &= \Phi_q(\omega_t^{\text{true}}, \Delta t) q_t^{\text{true}}, \\ \omega_{t+1}^{\text{true}} &= \omega_t^{\text{true}} + \Delta t J_{\text{true}}^{-1} [\hat{u}_t - (\omega_t^{\text{true}} \times J_{\text{true}} \omega_t^{\text{true}})]. \end{aligned} \quad (40)$$

Two execution modes are considered. In the perturbed open-loop rollout, the nominal control sequence is applied directly to the perturbed plant without feedback:

$$u_t^{\text{cmd}} = u_t^{\text{ref}}.$$

In the feedforward-PD tracking simulation, the measured state is used to compute the feedback tracking error.

The measured attitude and angular rate are modeled as

$$\hat{q}_t = \delta q_t \otimes q_t^{\text{true}}, \quad \hat{\omega}_t = \omega_t^{\text{true}} + \eta_{\omega,t}, \quad (41)$$

TABLE IX
Monte Carlo nominal planning performance over randomized initial attitudes.

Method	Trials	Success rate [%]	Mean e_q^f [deg]	Mean $\delta_F^{\max}$ [deg]	Mean runtime [s]
Proposed	500	98.4	0.1292	0.0	1.5415
SCP	500	97.8	0.0146	0.0746	4.6582
IPOPT	500	98.0	0.2993	0.0	0.4422

TABLE X
Solver benchmark summary for the Monte Carlo nominal planning.

Method	Mean iter.	Max iter.	Solver Fail.	Mean cost
Proposed	1228.276	1702	0	10.3361
SCP	9.17	13	0	11.0457
IPOPT	80.574	1000	8	10.6952

TABLE XI
Success thresholds for trajectory tracking and receding horizon MPC simulation.

Metric	Value	Unit	Metric	Value	Unit
ε_q	1.0	deg	ε_ω	2×10^{-2}	rad/s
ε_F	5×10^{-1}	deg	ε_b	1×10^{-6}	—

where δq_t represents the attitude measurement uncertainty and $\eta_{\omega,t}$ denotes the angular-rate measurement error.

$$\delta q_t = \left(\cos \frac{\epsilon_t}{2}, a_t \sin \frac{\epsilon_t}{2} \right), \quad \epsilon_t \sim \mathcal{N}(0, (0.05 \text{ deg})^2),$$

The commanded torque is computed as

$$u_t^{\text{cmd}} = \text{sat}_{u_{\max}} \left(u_t^{\text{ref}} + K_q e_{q,t} - K_\omega e_{\omega,t} \right).$$

The quaternion tracking error is defined using the measured attitude:

$$q_{e,t} = (\hat{q}_t)^{-1} \otimes q_t^{\text{ref}}, \quad q_{e,t} \leftarrow \text{sign}(q_{e,t,0}) q_{e,t},$$

and the attitude and angular-rate tracking errors are

$$e_{q,t} = q_{e,t,v}, \quad e_{\omega,t} = \hat{\omega}_t - \omega_t^{\text{ref}}.$$

The feedforward term retains the nominal optimized control structure, while the feedback terms compensate for trajectory deviations caused by inertia mismatch, actuator uncertainty, measurement error, and external disturbances.

Table XIII summarizes the trajectory tracking simulation results. The perturbed open-loop result shows a substantial degradation in terminal attitude accuracy, increasing the terminal attitude error from 0.1301 deg for the nominal reference to 4.2226 deg. This indicates that the nominal control sequence alone is sensitive to model and execution uncertainties when applied without feedback.

By contrast, the feedforward-PD tracking controller significantly reduces the terminal error. The terminal attitude error decreases to 0.4761 deg, and the RMS attitude tracking error is reduced from 2.5374 deg in the open-loop case to 0.2024 deg. The terminal angular-rate error is also reduced from 0.0088 to 0.0035, showing that the feedback terms compensate for the accumulated deviation caused by the perturbed plant.

All three trajectories satisfy the keep-out constraint, as indicated by the positive minimum angular margin and

TABLE XII
Uncertainty sources used in the perturbed-plant simulations.

Uncertainty source	Distribution / bound
Inertia mismatch, $\delta J_{,i}$	$\delta J_{,i} \sim \mathcal{U}[-5\%, 5\%]$
Actuator gain error, $\delta u_{,i}$	$\delta u_{,i} \sim \mathcal{U}[-2\%, 2\%]$
Torque bias, u_b	$u_b \sim \mathcal{U}[-0.01 u_{\max}, 0.01 u_{\max}]^3$
Disturbance torque, $\tau_{d,t}$	$\tau_{d,t} \sim \mathcal{N}(0, (0.01 u_{\max})^2 I_3)$
Measurement error (angle), ϵ_t	$\epsilon_t \sim \mathcal{N}(0, (0.05 \text{ deg})^2)$
Measurement error (rate), $n_{\omega,t}$	$n_{\omega,t} \sim \mathcal{N}(0, (10^{-4})^2 I_3)$
Control torque saturation	$\hat{u}_t = \text{sat}_{u_{\max}}(\cdot)$

zero maximum violation. Therefore, the tracking simulation shows that the feedforward-PD controller improves execution accuracy under uncertainty while preserving constraint satisfaction for this representative scenario.

E. Receding-horizon MPC simulation

Closed-loop replanning performance is evaluated in a receding-horizon MPC setting under the same perturbation model used in the trajectory tracking simulation. In contrast to the fixed-reference tracking rollout, the optimizer is repeatedly reinitialized with the current measured state and resolves the constrained trajectory optimization problem online. This simulation is performed as a Monte Carlo study over inertia uncertainty, actuator gain error, torque bias, disturbance torque, and measurement noise.

At guidance step k , the measurement model in (41) provides the measured state $\hat{x}_k = [\hat{q}_k^T \ \hat{\omega}_k^T]^T$. The optimizer uses the nominal inertia matrix J , while the plant is propagated with the perturbed inertia J_{true} according to (40). Thus, the true model parameters are not provided to the optimizer, and model mismatch is handled only through repeated state-feedback replanning.

A shrinking horizon is used to keep the final mission time fixed:

$$N_k = N - k.$$

At each step, the following nominal finite-horizon problem is solved:

$$\begin{aligned} & \underset{\{x_{i|k}, u_{i|k}\}}{\text{minimize}} && \sum_{i=0}^{N_k-1} \|u_{i|k}\|^2 + \lambda_N \|x_{N_k|k} - x_{\text{des}}\|^2 \\ & \text{subject to} && x_{0|k} = \hat{x}_k, \\ & && x_{i+1|k} = f_{\text{nom}}(x_{i|k}, u_{i|k}; J), \\ & && \|u_{i|k}\|_\infty \leq u_{\max}, \quad \|\omega_{i|k}\|_\infty \leq \omega_{\max}, \\ & && m_F(q_{i|k}) \geq 0, \\ & && i = 0, \dots, N_k - 1. \end{aligned} \quad (42)$$

TABLE XIII
Trajectory tracking simulation results.

Execution mode	e_q^f [deg]	e_ω^f	RMS e_q [deg]	Min m_F [deg]	$\delta_F^{\max}$ [deg]
Nominal reference	0.1301	0.0047	—	0.0013	0.0
Perturbed open-loop	4.2226	0.0088	2.5374	0.4211	0.0
Feedforward-PD tracking	0.4761	0.0035	0.2024	0.0484	0.0

For solvers that impose the terminal condition as a hard constraint, the terminal penalty terms in (42) are replaced by the corresponding equality constraints.

Only the first optimized input is applied to the plant:

$$u_k^{\text{cmd}} = u_{0|k}^*.$$

The applied torque is then generated by the perturbed actuator model in (39), and the resulting true state is used at the next guidance step. The previous predicted trajectory is shifted and used as the warm start for the next solve. If the previous solution is unavailable or dimensionally incompatible, a PD-generated trajectory is used as the initial guess.

For fairness, the same sampled perturbation realization is used for the proposed method, SCP, and IPOPT in each Monte Carlo trial. All methods use the same initial state, nominal model, mission scenario, constraints, horizon schedule, and warm-start rule. The MPC results therefore evaluate both closed-loop trajectory quality and suitability for repeated online replanning.

Table XIV summarizes the Monte Carlo MPC performance. The proposed method achieves the highest success rate of 97%, compared with 50% for SCP and 88% for IPOPT. Although SCP gives the smallest mean terminal attitude error, 0.3352 deg, its success rate is significantly lower and its mean solve time per step is the largest at 1.7329 s. IPOPT has the shortest mean solve time, 0.2398 s, but its success rate is lower than that of the proposed method and it exhibits more solver failures.

Figure 15 further examines the per-step solve-time behavior. The proposed method remains below the update-cycle limit over the full horizon, and its solve time decreases as the shrinking horizon becomes shorter. SCP shows the largest timing variability and exceeds the update-cycle limit at the initial guidance step, indicating that repeated online replanning can be restrictive for short update periods. IPOPT is faster at later guidance steps, but this timing advantage should be interpreted together with its lower success rate and larger failure count.

The proposed method shows favorable closed-loop reliability in this MPC setting. It achieves a mean terminal attitude error of 0.3591 deg and a mean maximum keep-out violation of 0.0431 deg, while reducing the mean solve time per step by more than half relative to SCP. As shown in Table XV, the proposed method also gives the lowest mean cost, 10.3798, with three failed trials.

In contrast, IPOPT records ten failures and SCP, despite having no solver failure count, produces a substantially lower closed-loop success rate. These results indicate that the proposed solver offers a favorable balance between

Algorithm 1 Monte Carlo receding-horizon robustness simulation

- 1: Sample J_{true} , D_u , u_b , disturbance sequences, and measurement-noise sequences.
- 2: Initialize $x_0^{\text{true}} = (q_0, \omega_0)$.
- 3: **for** $k = 0, \dots, N - 1$ **do**
- 4: Measure $\hat{x}_k = (\hat{q}_k, \hat{\omega}_k)$ using (41).
- 5: Shift the previous prediction to construct a warm start.
- 6: Solve the nominal receding-horizon OCP using Proposed, SCP, or IPOPT.
- 7: Apply only the first input $u_{0|k}^*$ through (39).
- 8: Propagate the perturbed true plant using (40).
- 9: **end for**
- 10: Compute terminal errors, keep-out margins, bound violations, failure counts, and timing statistics.

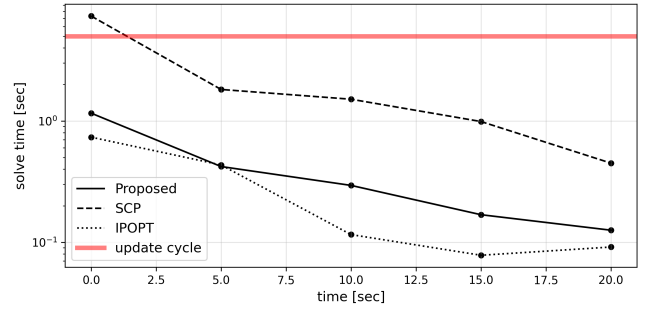


Fig. 15. Per-step solve-time distribution in the receding-horizon MPC simulation. The plot illustrates timing variability and update cycle satisfaction.

trajectory quality, constraint satisfaction, and timing consistency for receding-horizon implementation.

This MPC study is empirical and does not constitute a formal proof of recursive feasibility or robust invariance. Such guarantees would require additional terminal ingredients, constraint tightening, and invariant-set analysis, which are outside the scope of this work.

F. Computational Complexity and Horizon-Length Scaling

Although the proposed solver avoids solving a coupled horizon-wide nonlinear program or a sequence of convexified trajectory-optimization subproblems inside each iteration, the dynamics graph projection requires small nonlinear least-squares solves. We therefore explicitly profile the computational cost of the LM dynamics

TABLE XIV
Monte Carlo receding-horizon MPC performance under model and actuator uncertainty.

Method	Trials	Success [%]	Mean e_q^f [deg]	Mean $\delta_F^{\max}$ [deg]	Mean time/step [s]
Proposed	100	97	0.3591	0.0431	0.7408
SCP	100	50	0.3352	0.0414	1.7329
IPOPT	100	88	0.4017	0.0404	0.2398

TABLE XV
Solver benchmark summary for receding-horizon implementation.

Method	Mean iter.	Max iter.	Solver fail.	Mean cost
Proposed	2438.78	4397	3	10.3798
SCP	21.10	24	0	10.9911
IPOPT	244.11	1733	10	11.0315

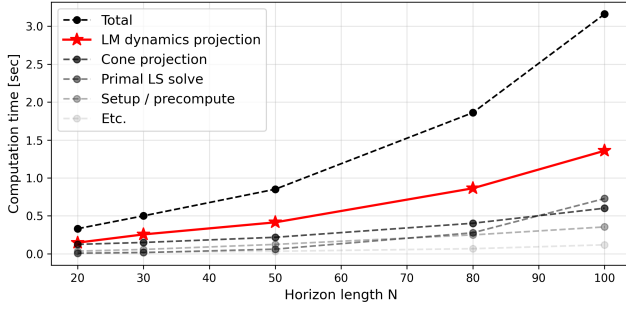


Fig. 16. ADMM computation time for varying N .

projection. The constrained fly-by scenario is fixed, and only the horizon length is varied over

$$N \in \{20, 30, 50, 80, 100\}.$$

The mission geometry, boundary conditions, cone constraints, penalty parameters, and stopping tolerances are kept identical to Scenario 2 throughout the sweep. Let K_{ADMM} denote the number of ADMM iterations, $n_x = 7$ the dimension of the local state $x_t = (q_t, \omega_t)$, and $\bar{\kappa}_{\text{LM}}$ the average number of LM inner iterations per time node. For one LM inner iteration, the dominant operations are the evaluation of f_{exp} , the evaluation of its Jacobian, the assembly of the normal equations, and the solution of a damped $n_x \times n_x$ linear system. Hence, the serial cost of the dynamics graph projection per ADMM iteration is

$$\mathcal{O}(N\bar{\kappa}_{\text{LM}}(C_f + C_J + n_x^3)),$$

where C_f and C_J denote the costs of evaluating f_{exp} and its Jacobian, respectively. Since $n_x = 7$ is fixed for the attitude problem, this scales as $\mathcal{O}(N\bar{\kappa}_{\text{LM}})$ for fixed spacecraft state dimension. The box projections and dual updates scale as $\mathcal{O}(N)$, while the attitude cone projection scales as $\mathcal{O}(Nn_c n_{\text{sweep}})$ for a fixed number of cone constraints n_c and cyclic projection sweeps n_{sweep} .

The timing results in Fig. 16 and Table XVI indicate that the overall computational cost increases smoothly as the horizon length is enlarged. Among the individual projection operators, the LM dynamics projection accounts

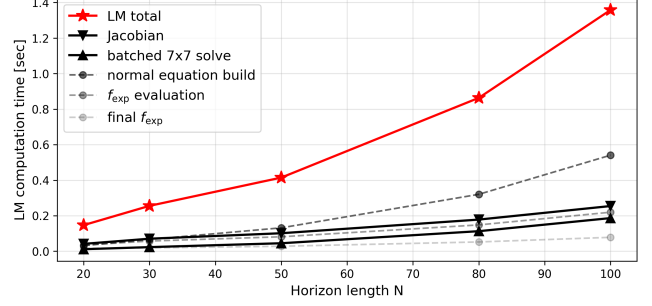


Fig. 17. LM computation time for varying N .

for the largest fraction of the projection time. However, its runtime exhibits a regular scaling trend with respect to N , rather than a rapid growth that would indicate increasing numerical difficulty of the local dynamics projection.

Figure 17 provides a more detailed decomposition of the LM dynamics-projection cost into f_{exp} evaluation, Jacobian evaluation, normal-equation assembly, batched 7×7 linear solves, and final residual evaluation. The decomposition shows that the fixed-size linear algebra kernel is not the dominant bottleneck. Instead, the computational cost is mainly determined by residual and Jacobian evaluations and by the construction of the normal equations across the time nodes. This is consistent with the structure of the proposed projection, where each local LM update involves only a fixed-dimensional linear solve, while the number of independent projection blocks grows with the horizon length.

The LM iteration statistics in Table XVII further show that the local dynamics projections remain well conditioned over the tested horizon range. The average number of LM iterations remains small, and the maximum iteration count does not exceed three even for $N = 100$. Moreover, the reduced LM projection objective stays close to numerical precision, indicating that the local graph projections are solved accurately. The solution-quality metrics in Table XVIII also remain stable across the tested horizons, with comparable control effort, small terminal error, and negligible keep-in and keep-out violations.

These results suggest that the increase in total runtime is primarily attributable to the larger number of time nodes and the corresponding increase in ADMM-level iterations, rather than to a degradation in the conditioning or convergence behavior of each local LM dynamics projection. Therefore, the horizon-length scaling behavior is mainly driven by the growth in the number of local

TABLE XVI
Horizon-length scaling and normalized timing of the proposed solver.

N	K_{ADMM}	Total [s]	Setup [s]	Primal update [s]	LM [s]	Cone projection [s]	Others [s]	Memory usage [MB]
20	387	0.4130	0.0416	0.0115	0.1815	0.1550	0.0235	0.4674
30	446	0.6242	0.0638	0.0231	0.3203	0.1869	0.0301	0.9976
50	622	0.9027	0.1384	0.0674	0.4285	0.2307	0.0377	2.6486
80	1057	1.9798	0.2893	0.3012	0.8924	0.4201	0.0766	6.6014
100	1469	3.0935	0.3748	0.6573	1.3465	0.5978	0.1171	10.2208

TABLE XVII
LM dynamics-projection iteration statistics.

N	Mean LM iter.	Max LM iter.	LM objective
20	1.8413	3	1.55×10^{-7}
30	1.8401	3	2.35×10^{-9}
50	1.8728	2	4.41×10^{-9}
80	1.9157	2	1.71×10^{-9}
100	1.9159	2	9.62×10^{-10}

TABLE XVIII
Solution quality in the horizon-length scaling study.

N	$\ u\ ^2$	$\ x_N - x_{\text{des}}\ $	Keep-out violation [deg]	Keep-in violation [deg]
20	14.9730	0.0038	0.0039	0.0010
30	16.0311	0.0129	0.0016	0.0016
50	15.7673	0.0215	0.0	0.0049
80	15.1836	0.0262	0.0	0.0002
100	14.8363	0.0281	0.0	0.0

projection blocks, while the per-block LM projection remains computationally lightweight and numerically reliable.

G. Hyperparameter Sensitivity

The convergence discussion in Section IV-H provides a local stationary-convergence result for an inexact ADMM scheme under regularity, boundedness, Kurdyka-Łojasiewicz, and summable projection-error assumptions. However, the practical implementation uses fixed projection and ADMM tolerances, and the projection operators include nonconvex attitude cone projections and local Levenberg-Marquardt graph projections. Therefore, the numerical behavior can still depend on the penalty parameter, terminal weight, projection accuracy, and quality of the local projection steps.

For this reason, we complement the conditional convergence analysis with an empirical hyperparameter sensitivity study. The objective is to evaluate whether the proposed method exhibits stable convergence and feasibility over a range of practically relevant parameter choices, rather than only at a single manually tuned setting. The same Scenario 2 setup, initialization, stopping criteria, and feasibility checks are used throughout the sweep.

To better understand the empirical behavior of both the proposed and baseline methods, we perform a parameter sensitivity analysis over key hyperparameters.

- *Proposed Method:* Our method is based on a non-convex ADMM framework using augmented Lagrangians. We sweep the penalty parameter ρ as well as the terminal cost weight λ_N (penalizing deviation from q_{des} and ω_{des}) to study their effects on convergence speed and feasibility.
- *Baseline (IPOPT):* The baseline uses an IPOPT scheme implemented via CasADi. For this comparison, we sweep the control-effort weight λ_u and terminal cost weight λ_N to examine the sensitivity of the direct NLP baseline to objective weighting and penalty tuning.

The SCP baseline is excluded from this hyperparameter sensitivity comparison because it solves a penalty-relaxed formulation of the constraints, which makes direct comparison less consistent with the proposed method and IPOPT.

Figure 18 compares the two methods across their respective hyperparameter grids. Green regions denote feasible solutions (*i.e.*, those that satisfy terminal and intermediate constraints), while the red cell highlights the best-performing run in terms of computation time and terminal error. The results show that the proposed method exhibits a wider feasible-convergence region over the tested (ρ, λ_N) grid, indicating lower sensitivity to hyperparameter selection. The best-performing configuration of the proposed method achieves a computation time of 0.954 s, whereas the IPOPT baseline reaches its best case at 0.213 s. Although IPOPT can be faster for carefully tuned penalty weights, its feasible region is more fragmented and several parameter combinations lead to substantially larger runtimes. The proposed method shows more uniform timing behavior across the feasible region, suggesting more robust convergence with respect to penalty and ADMM parameter variations.

These results provide empirical evidence that the proposed solver is less sensitive to the tested ADMM penalty and terminal-weight parameters than the direct NLP baseline in this scenario. This supports its use as a candidate method for constrained online replanning.

VI. CONCLUDING REMARKS

This paper presented a projection-based first-order optimization framework for spacecraft attitude maneuver planning under nonlinear dynamics and nonconvex orientation constraints. The problem was reformulated using linear dynamics augmented with auxiliary variables, and

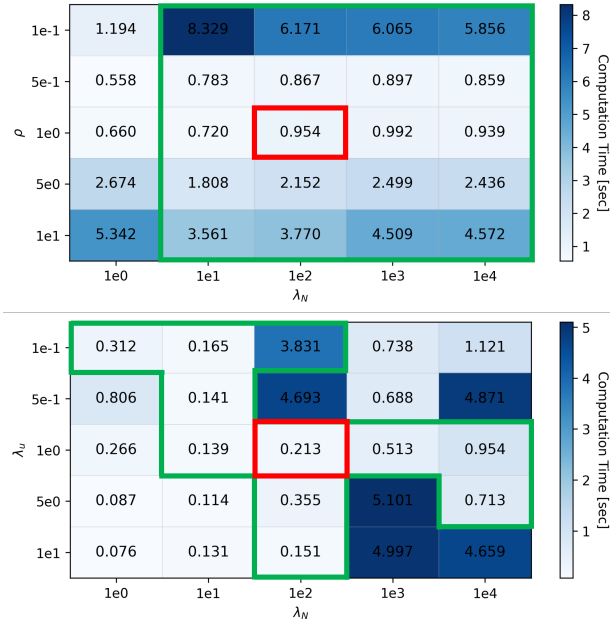


Fig. 18. Effect of hyperparameters on convergence performance. Green cells indicate feasible solutions; red denotes best-performing configuration. Proposed method (top) vs. IPOPT baseline (bottom).

constraints were enforced via modular projections within an ADMM loop.

Nonconvex constraints were handled using either direct geometric projections (for orientation cone constraints) or iterative projections (for nonlinear dynamics constraints) onto the corresponding nonconvex sets. This avoided repeated linearization, improving computational efficiency.

In addition, we provided a conditional local stationarity analysis for the proposed nonconvex ADMM scheme. The analysis characterized the approximation errors of the multi-cone attitude projection and the LM dynamics graph projection using residual-based local error bounds. Under regularity, boundedness, Kurdyka–Łojasiewicz, and summable projection-error assumptions, the resulting inexact ADMM iterates were shown to approach a stationary point of the split problem.

Simulations on agile maneuver scenarios demonstrated that the proposed method achieves competitive terminal accuracy and control cost relative to the baseline methods (SCP and IPOPT). In scenarios with keep-in and keep-out cones, the method successfully kept targets within the camera’s boresight while avoiding bright object exposure.

Monte Carlo simulations further demonstrated that the proposed method provides reliable trajectory generation and repeated replanning performance compared with the baselines. In randomized initial-condition tests and receding-horizon replanning scenarios, the proposed solver achieved the highest success rate while maintaining terminal accuracy and pointing-constraint violations within the prescribed tolerances. Trajectory tracking and MPC simulations under inertia mismatch, actuator uncertainty, disturbance torque, and measurement noise indicate that the proposed framework may serve as a useful basis for

robust constrained attitude guidance in uncertain operating conditions.

Further analysis showed that the proposed method is substantially less sensitive to the tested hyperparameter variations, although IPOPT remains faster in average runtime when it converges. These results suggest that the proposed framework is a promising basis for constrained attitude guidance and online replanning, while embedded real-time validation remains future work.

Overall, the method provides a modular and scalable projection-based approach with competitive trajectory quality and reliable constraint satisfaction in the tested scenarios.

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During the preparation of this work, the authors used ChatGPT by OpenAI to check grammar, spelling, phrasing, and equation formatting, without changing the technical content. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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